

JESS Thermodynamic Database v8.9

Gold

24-Jan-23

Reaction No. 4707							
$\text{Au+3_OH-1(4)} + 2\text{<H+1>} + 2\text{<Cl-1>} = \text{Au+3_Cl-1(2)_OH-1(2)} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.57(4SF)	0	CRV	2556

Reaction No. 4708							
$\text{Au+3_OH-1(4)} + 3\text{<H+1>} + 3\text{<Cl-1>} = \text{Au+3_Cl-1(3)_OH-1} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:23.6(3SF)	0	CRV	2556

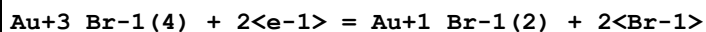
Reaction No. 4710							
$\text{Au+3_OH-1(4)} + \text{OH-1} = \text{Au+3_OH-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.6(1SF)	4	CRV	2556

Reaction No. 4711							
$\text{Au+3_OH-1(4)} + 4\text{<H+1>} + 4\text{<Cl-1>} = \text{Au+3_Cl-1(4)} + 4\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:29.6(3SF)	0	CRV	2556

Reaction No. 4721							
$\text{Au+3_Cl-1(4)} + 2\text{<e-1>} = \text{Au+1_Cl-1(2)} + 2\text{<Cl-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:31.2(0.1SD)	0	CMN	886
3	25	0	Inf. Dilution	E:0.9251(4SF)	0	CRV	
4	25	0	Inf. Dilution	E:1.15(0.01SD)	0	MEF	775
5	25	0	Inf. Dilution	E:0.926(3SF)	0	CRV	3006
6	25	0	Inf. Dilution	E:0.926(3SF)	0	CRV	22551

7	25	0.2:2	HCl	E:0.926(3SF)	0	MEF	778
---	----	-------	-----	--------------	---	-----	-----

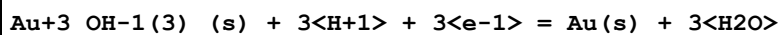
Reaction No. 4722



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.80(0.01SD)	0	MEF	775
3	25	0	Inf. Dilution	E:0.802(3SF)	0	CRV	3006
4	25	0	Inf. Dilution	E:0.802(3SF)	0	CRV	22551
5	25	0.1:1	HBr	E:0.802(3SF)	0	MEF	778

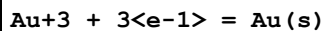
Note (5): reactive medium

Reaction No. 4723



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:69.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.45(0.01SD)	0	MDG	775
3	25	0	Inf. Dilution	E:1.45(3SF)	0	CRV	6330

Reaction No. 4724



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:76.0(3SF)	4	ENS	140
2	25	0	Inf. Dilution	dG:-105.2(0.1SD)	0	ENS	5323
3	25	0	Inf. Dilution	dG:-103.6(4SF)	3	CRV	6488
4	25	0	Inf. Dilution	dG:-433.(3SF) kJ	5	CRV	7421
5	25	0	Inf. Dilution	dG:-103.600(6SF)	4	CRV	8875
6	25	0	Inf. Dilution	dG:-103.600(6SF)	4	CRV	9029
7	25	0	Inf. Dilution	dG:-103.60(5SF)	4	CRV	10174
8	25	0	Inf. Dilution	dG:-103.60(5SF)	3	EOD	31585
9	50	0	Inf. Dilution	dG:-105.08(5SF)	2	CRV	10174
10	75	0	Inf. Dilution	dG:-106.62(5SF)	2	CRV	10174
11	100	0	Inf. Dilution	dG:-108.21(5SF)	2	CRV	10174
12	125	0	Inf. Dilution	dG:-109.85(5SF)	2	CRV	10174
13	150	0	Inf. Dilution	dG:-111.55(5SF)	3	CRV	10174
14	175	0	Inf. Dilution	dG:-113.31(5SF)	2	CRV	10174
15	200	0	Inf. Dilution	dG:-115.15(5SF)	0	CRV	10174

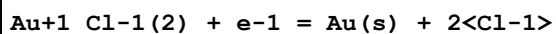
16	225	0	Inf. Dilution	dG:-117.06 (5SF)	0	CRV	10174
17	250	0	Inf. Dilution	dG:-119.08 (5SF)	0	CRV	10174
18	300	0	Inf. Dilution	dG:-123.44 (5SF)	0	CRV	10174
19	25	0	Inf. Dilution	E:1.498 (4SF)	5	MEF	778
20	25	0	Inf. Dilution	E:1.52 (3SF)	0	CRV	3006
21	25	0	Inf. Dilution	E:1.498 (4SF)	5	CRV	6330
22	25	0	Inf. Dilution	E:1.50 (3SF)	4	CRV	7372
23	25	0	Inf. Dilution	E:1.50 (3SF)	4	CRV	7761
24	25	0	Inf. Dilution	E:1.52 (3SF)	0	CRV	22551
25	25	1	H2SO4/m	E:1.362 (4SF)	4	CRV	22551
26	25	0	Inf. Dilution	dH:-96.93 (4SF)	2	CRV	6488
27	25	0	Inf. Dilution	dH:-98.913 (5SF)	3	CRV	8875
28	25	0	Inf. Dilution	dH:-97.800 (5SF)	3	CRV	9029
29	25	0	Inf. Dilution	dH:-97.800 (5SF)	3	EOD	31585
30	25	0	GAMSPATH	dH:-413.850 (6SF) kJ	3	CRV	12638

Reaction No. 4725							
Au+3_Cl-1(4) + 3<e-1> = Au(s) + 4<Cl-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.6 (0.1SD)	5	CMN	886
2	25	0	Inf. Dilution	E:0.994 (3SF)	5	CRV	
3	25	0	Inf. Dilution	E:1.00 (0.01SD)	4	MEF	775
4	25	0	Inf. Dilution	E:1.002 (4SF)	3	CRV	3006
5	25	0	Inf. Dilution	E:1.002 (4SF)	3	CRV	6330
6	25	0	Inf. Dilution	E:1.00 (3SF)	4	ENS	7276
7	25	0	Inf. Dilution	E:1.002 (4SF)	3	CRV	22551
8	25	0.2:2	HCl	E:1.002 (4SF)	3	MEF	778
9	25	0	Inf. Dilution	dH:-87.0 (0.1SD)	5	MDG	892

Reaction No. 4726							
Au+3_Br-1(4) + 3<e-1> = Au(s) + 4<Br-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.3	NaClO4	lgK:41. (2SF)	3	EBP	23202
Note (1): 4<Br-1> + Au+3 = Au+3_Br-1(4) good datum ruined by Au+3 + 3<e-1> > < f(I)							
2	25	0	Inf. Dilution	E:0.85 (0.01SD)	5	MEF	775
3	25	0	Inf. Dilution	E:0.854 (3SF)	5	CRV	3006

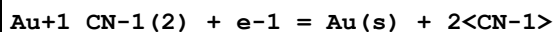
4	25	0	Inf. Dilution	E:0.854 (3SF)	5	CRV	6330
5	25	0	Inf. Dilution	E:0.854 (3SF)	4	CRV	22551

Reaction No. 4727



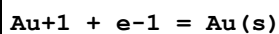
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:19.1 (0.05SD)	0	MEF	885
2	25	0	Inf. Dilution	lgK:19.3 (3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:19.5 (0.05SD)	0	CMN	887
4	25	0	Inf. Dilution	E:1.113 (4SF)	0	CRV	
5	25	0	Inf. Dilution	E:1.15 (0.01SD)	0	MEF	775
6	25	0	Inf. Dilution	E:1.154 (4SF)	0	CRV	3006
7	25	0	Inf. Dilution	E:1.154 (4SF)	0	CRV	22551
8	25	0.2:2	HCl	E:1.1540 (5SF)	0	MEF	778
9	25	0.34	HCl	dH:-32.2 (0.1SD)	0	MEF	888

Reaction No. 4728



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:-10. (2SF)	0	ENS	775
2	25	0	Inf. Dilution	lgK:-9.8 (0.1SD)	5	ENS	5569
3	25	0	Inf. Dilution	E:-0.669 (3SF)	0	MEF	778
4	25	0	Inf. Dilution	E:0.6 (1SF)	0	ENS	7276
5	25	0.5	KCN	E:-0.595 (3SF)	4	CRV	22551

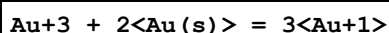
Reaction No. 5085



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.6 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:28.6 (0.05SD)	2	ENS	802
3	25	0.025	KCN	lgK:28. (0.4SD)	0	ENS	884
4	25	0	Inf. Dilution	dG:-39.00 (4SF)	0	CRV	6488
5	25	0	Inf. Dilution	dG:-176. (3SF) kJ	3	CRV	7421
6	25	0	Inf. Dilution	dG:-39.000 (5SF)	0	CRV	8875
7	25	0	Inf. Dilution	dG:-39.000 (5SF)	0	CRV	9029
8	25	0	Inf. Dilution	dG:-39.00 (4SF)	0	CRV	10174
9	50	0	Inf. Dilution	dG:-38.39 (4SF)	1	CRV	10174

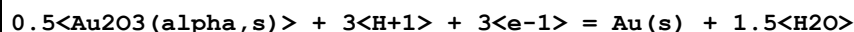
10	75	0	Inf. Dilution	dG:-37.77 (4SF)	1	CRV	10174
11	100	0	Inf. Dilution	dG:-37.14 (4SF)	1	CRV	10174
12	125	0	Inf. Dilution	dG:-36.50 (4SF)	1	CRV	10174
13	150	0	Inf. Dilution	dG:-35.86 (4SF)	1	CRV	10174
14	175	0	Inf. Dilution	dG:-35.22 (4SF)	1	CRV	10174
15	200	0	Inf. Dilution	dG:-34.57 (4SF)	1	CRV	10174
16	225	0	Inf. Dilution	dG:-33.92 (4SF)	0	CRV	10174
17	250	0	Inf. Dilution	dG:-33.27 (4SF)	0	CRV	10174
18	300	0	Inf. Dilution	dG:-31.99 (4SF)	0	CRV	10174
19	25	0	Inf. Dilution	E:1.692 (4SF)	0	MEF	778
20	25	0	Inf. Dilution	E:1.83 (3SF)	3	CRV	3006
21	25	0	Inf. Dilution	E:1.692 (4SF)	0	CRV	6330
22	25	0	Inf. Dilution	E:1.692 (4SF)	0	ENS	7381
23	25	0	Inf. Dilution	E:1.69 (3SF)	0	CRV	7761
24	25	0	Inf. Dilution	E:1.83 (3SF)	3	CRV	22551
25	25	0	Inf. Dilution	dH:-47.58 (4SF)	2	CRV	6488
26	25	0	Inf. Dilution	dH:-194.7 (4SF) kJ	2	CRV	7761
27	25	0	Inf. Dilution	dH:-48.175 (5SF)	2	CRV	8875
28	25	0	Inf. Dilution	dH:-47.900 (5SF)	2	CRV	9029
29	25	0	GAMSPATH	dH:-201.560 (6SF) kJ	3	CRV	12638

Reaction No. 5086



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-9.8 (0.07SD)	0	ENS	802
3	25	0	Inf. Dilution	lgK:-10. (2SF)	0	ENS	7276
4	50	0	Inf. Dilution	lgK:-7.9 (0.05SD)	0	ENS	890
5	60	0	Inf. Dilution	lgK:-7.2 (0.05SD)	0	ENS	890
6	100	0	Inf. Dilution	lgK:-4.8 (0.05SD)	0	ENS	890

Reaction No. 5087



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0	Inf. Dilution	lgK:68.1 (0.05SD)	0	MEF	890
2	25	0	Inf. Dilution	lgK:69.1 (0.05SD)	5	MEF	891

Reaction No. 5088							
$\text{Au}+3\text{Cl-1(4)} + 2\text{Au(s)} + 2\text{Cl-1} = 3\text{Au+1Cl-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.7(0.05SD)	5	CMN	886
Note (1): In reasonable accord with data of Gammons et al, 1997 [9031]							
2	25	0	Inf. Dilution	lgK:-7.39(3SF)	4	EIE	9031
3	50	0	Inf. Dilution	lgK:-6.33(3SF)	4	EIE	9031
4	100	0	Inf. Dilution	lgK:-4.42(0.22SD)	6	MSL	9031
Note (4): Excellent work spoiled only by correction to I=0							
5	150	0	Inf. Dilution	lgK:-2.86(0.12SD)	6	MSL	9031
6	200	0	Inf. Dilution	lgK:-1.45(0.19SD)	6	MSL	9031
7	25	0	Inf. Dilution	dH:117.9(4SF)kJ	0	EOD	9031
Note (7): Data from Helgeson, 1969 [5] not in accord with present study							
8	25	0	Inf. Dilution	dH:77.8(3SF)kJ	4	EOD	9031
Note (8): Data from Nikolaeva, 1972 [9034] in accord with present study							

Reaction No. 5089							
$2\text{Cl-1} + \text{Au+1} = \text{Au+1Cl-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	100	0	P=500 Inf. Dilution	lgK:7.70(3SF)	0	EOD	6495
2	150	0	P=500 Inf. Dilution	lgK:7.05(3SF)	0	EOD	6495
3	200	0	P=500 Inf. Dilution	lgK:6.65(3SF)	0	EOD	6495
4	250	0	P=500 Inf. Dilution	lgK:6.45(3SF)	0	EOD	6495
5	300	0	P=500 Inf. Dilution	lgK:6.44(3SF)	0	EOD	6495
6	100	0	P=1000 Inf. Dilution	lgK:7.56(3SF)	0	EOD	6495
7	150	0	P=1000 Inf. Dilution	lgK:6.90(3SF)	0	EOD	6495
8	200	0	P=1000 Inf. Dilution	lgK:6.47(3SF)	0	EOD	6495
9	250	0	P=1000 Inf. Dilution	lgK:6.22(3SF)	0	EOD	6495
10	300	0	P=1000 Inf. Dilution	lgK:6.11(3SF)	0	EOD	6495
11	100	0	P=2000 Inf. Dilution	lgK:7.34(3SF)	0	EOD	6495

12	150	0	P=2000 Inf. Dilution	lgK:6.66 (3SF)	0	EOD	6495
13	200	0	P=2000 Inf. Dilution	lgK:6.19 (3SF)	0	EOD	6495
14	250	0	P=2000 Inf. Dilution	lgK:5.89 (3SF)	0	EOD	6495
15	300	0	P=2000 Inf. Dilution	lgK:5.70 (3SF)	0	EOD	6495
16	100	0	P=3000 Inf. Dilution	lgK:7.18 (3SF)	0	EOD	6495
17	150	0	P=3000 Inf. Dilution	lgK:6.48 (3SF)	0	EOD	6495
18	200	0	P=3000 Inf. Dilution	lgK:5.99 (3SF)	0	EOD	6495
19	250	0	P=3000 Inf. Dilution	lgK:5.65 (3SF)	0	EOD	6495
20	300	0	P=3000 Inf. Dilution	lgK:5.42 (3SF)	0	EOD	6495
21	100	0	P=4000 Inf. Dilution	lgK:7.06 (3SF)	0	EOD	6495
22	150	0	P=4000 Inf. Dilution	lgK:6.34 (3SF)	0	EOD	6495
23	200	0	P=4000 Inf. Dilution	lgK:5.83 (3SF)	0	EOD	6495
24	250	0	P=4000 Inf. Dilution	lgK:5.47 (3SF)	0	EOD	6495
25	300	0	P=4000 Inf. Dilution	lgK:5.21 (3SF)	0	EOD	6495
26	100	0	P=5000 Inf. Dilution	lgK:6.97 (3SF)	0	EOD	6495
27	150	0	P=5000 Inf. Dilution	lgK:6.23 (3SF)	0	EOD	6495
28	200	0	P=5000 Inf. Dilution	lgK:5.71 (3SF)	0	EOD	6495
29	250	0	P=5000 Inf. Dilution	lgK:5.32 (3SF)	0	EOD	6495
30	300	0	P=5000 Inf. Dilution	lgK:5.04 (3SF)	0	EOD	6495
31	0	0	Inf. Dilution	lgK:10.4 (3SF)	0	EOD	6495
32	22			lgK:17. (2SF)	0	EES	140
33	25	0	Inf. Dilution	lgK:9. (1SF)	0	ENS	
34	25	0	Inf. Dilution	lgK:11.1 (3SF)	1	ELC	
35	25	0	Inf. Dilution	lgK:9.55 (3SF)	0	EOD	6495
36	25			lgK:9. (1SF)	0	EES	140

37	50	0	Inf. Dilution	lgK:8.4(0.05SD)	0	ENS	5022
38	50	0	Inf. Dilution	lgK:8.87(3SF)	0	EOD	6495
39	75	0	Inf. Dilution	lgK:8.32(3SF)	0	EOD	6495
40	100	0	Inf. Dilution	lgK:7.5(0.05SD)	0	ENS	5022
41	100	0	Inf. Dilution	lgK:7.88(3SF)	0	EOD	6495
42	125	0	Inf. Dilution	lgK:7.52(3SF)	0	EOD	6495
43	150	0	Inf. Dilution	lgK:6.9(2SF)	0	EES	5398
44	150	0	Inf. Dilution	lgK:7.24(3SF)	0	EOD	6495
45	175	0	Inf. Dilution	lgK:7.03(3SF)	0	EOD	6495
46	200	0	Inf. Dilution	lgK:6.88(3SF)	0	EOD	6495
47	225	0	Inf. Dilution	lgK:6.78(3SF)	0	EOD	6495
48	250	0	Inf. Dilution	lgK:6.75(3SF)	0	EOD	6495
49	275	0	Inf. Dilution	lgK:6.79(3SF)	0	EOD	6495
50	300	0	Inf. Dilution	lgK:6.92(3SF)	0	EOD	6495
51	50:100	0	Inf. Dilution	dH:-10.(0.8SD)	0	EVH	
52	25	0.1	DMSO Et4NC1O4	lgK:18.00(4SF)	0	MIE	5398
53	25	0.1	DMSO LiClO4	lgK:18.00(4SF)	0	MIE	5398
54	20	0.1	MeCN Et4NC1O4	lgK:20.2(3SF)	0	MEF	6758

Reaction No. 5090							
Au+3_Cl-1_OH-1(3) + H+1 + Cl-1 = Au+3_Cl-1(2)_OH-1(2) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:8.1(0.05SD)	0	MGL	895
2	25	0	Inf. Dilution	lgK:6.7(2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:8.0(0.05SD)	0	MGL	893

Reaction No. 5091							
Au+3_OH-1(4) + H+1 + Cl-1 = Au+3_Cl-1_OH-1(3) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:8.5(0.05SD)	5	MGL	895
2	25	0	Inf. Dilution	lgK:8.7(0.05SD)	5	MGL	893

Reaction No. 5092							
Au+3_Cl-1(2)_OH-1(2) + H+1 + Cl-1 = Au+3_Cl-1(3)_OH-1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	20	0	Inf. Dilution	lgK:7.0(0.05SD)	5	MGL	895
2	25	0	Inf. Dilution	lgK:7.1(0.05SD)	5	MGL	893
3	25	3	NaClO4	lgK:7.0(0.05SD)	5	MGL	
Note (3): From: Acta Chem.Scand.,1967,21,819							

Reaction No. 5093							
Au+3_Cl-1(3)_OH-1 + H+1 + Cl-1 = Au+3_Cl-1(4) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:6.1(0.05SD)	5	MGL	895
2	21	0	Inf. Dilution	lgK:6.27(3SF)	5	MSP	5398
3	21	0.02	NaClO4	lgK:6.15(3SF)	5	MSP	5398
4	25	0	Inf. Dilution	lgK:6.2(0.05SD)	5	MGL	893
5	25	3	NaClO4	lgK:6.2(0.05SD)	5	MGL	
Note (5): From: Acta Chem.Scand.,1967,21,819							
6	50	0	Inf. Dilution	lgK:5.76(3SF)	5	MSP	5398
7	50	0.02	NaClO4	lgK:5.63(3SF)	5	MSP	5398

Reaction No. 5094							
Au+3_Cl-1(3) + H2O = Au+3_Cl-1(3)_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-1.0(0.3SD)	0	MPS	896
2	25	0	Inf. Dilution	lgK:-1.6(2SF)	1	ELC	
3	25	3	NaClO4	lgK:-2.7(0.05SD)	0	MGL	
Note (3): From: Acta Chem.Scand.,1967,21,819							

Reaction No. 5095							
4<Cl-1> + Au+3 = Au+3_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0	Inf. Dilution	lgK:21.30(4SF)	0	MEF	140
2	18	0	Inf. Dilution	lgK:26.(2.SD)	0	ENS	600
3	18			lgK:26.(2SF)	0	MEF	140
4	25	0	Inf. Dilution	lgK:25.5(3SF)	1	ELC	
5	25	0.3	NaClO4	lgK:25.2(3SF)	0	EOD	23202
6	50	0	Inf. Dilution	lgK:25.(0.5SD)	0	ENS	5
7	100	0	Inf. Dilution	lgK:22.(0.5SD)	0	ENS	5
8	150	0	Inf. Dilution	lgK:21.0(3SF)	0	EES	5398
9	18:100	0	Inf. Dilution	dH:-22.(1.SD)	0	CRV	2556

Reaction No. 5096							
Au+3_Cl-1(3) + Cl-1 = Au+3_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:4.9(0.2SD)	4	MSP	896
2	25	0	Inf. Dilution	lgK:4.34(3SF)	4	MEF	931
3	25	2	HClO4	lgK:5.0(0.05SD)	5	MKN	897

Reaction No. 5097							
Au+3_OH-1(4) + H2O = Au+3_OH-1(5) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.4(0.05SD)	4	ENS	4998

Reaction No. 5098							
Au+3_OH-1(3) + OH-1 = Au+3_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.3(0.05SD)	0	CSM	8

Reaction No. 5099							
Au+3_OH-1(3)_(s) = Au+3_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.5(0.05SD)	0	ENS	4998

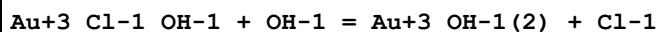
Reaction No. 5100							
Au+3_OH-1(5) - 2<H2O> = AuO3-3 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.0(0.2SD)	0	ENS	4998

Reaction No. 5101							
Au+3_OH-1 + H2O = Au+3_OH-1(2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.5(0.2SD)	3	ENS	4998

Reaction No. 5102							
Au+3_OH-1(3) + 3<H+1> + 3<e-1> = Au(s) + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

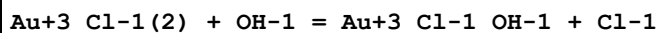
1	25	0	Inf. Dilution	lgK:73.(0.5SD)	0	ENS	802
---	----	---	---------------	----------------	---	-----	-----

Reaction No. 5103



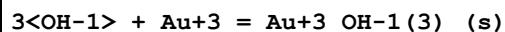
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01:0.5	NaCl	lgK:6.0(0.2SD)	3	MEF	5466

Reaction No. 5104



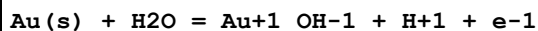
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01:0.5	Unknown	lgK:6.6(0.2SD)	3	MEF	5466

Reaction No. 5105



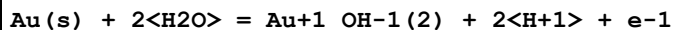
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	0.45	HNO3	lgK:45.3(0.05SD)	0	MSL	898
2	25	0	Inf. Dilution	lgK:48.91(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:48.9(3SF)	1	EOR	
4	25	0	Inf. Dilution	lgK:47.5(3SF)	0	ENS	773
5	25	0	Inf. Dilution	lgK:-53.(2SF)	0	CRV	22551
6	25	1	Unknown	lgK:46.9(3SF)	2	ENS	773

Reaction No. 5115



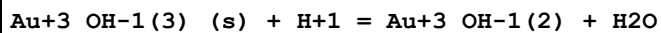
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-24.3(3SF)	1	ELC	
2	25	0	NaOH	lgK:-23.(0.5SD)	0	MSL	900

Reaction No. 5116



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.4(3SF)	1	ELC	
2	25	0	NaOH	lgK:-35.(1.SD)	0	ENS	900

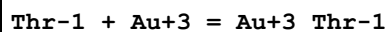
Reaction No. 5120



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

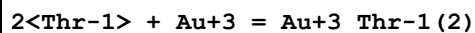
1	18	1	H2SO4	lgK:-4.(0.7SD)	0	ENS	4998
---	----	---	-------	----------------	---	-----	------

Reaction No. 5589



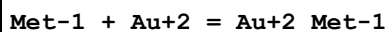
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:6.74(3SF)	4	MGL	905

Reaction No. 5590



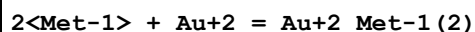
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:10.19(4SF)	4	MGL	905

Reaction No. 5684



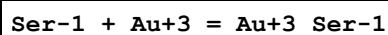
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:7.23(3SF)	6	MGL	905

Reaction No. 5685



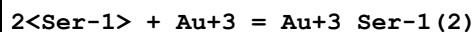
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:10.19(4SF)	6	MGL	905

Reaction No. 5716



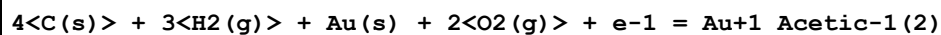
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:6.54(3SF)	3	MGL	905

Reaction No. 5717



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:9.34(3SF)	3	MGL	905

Reaction No. 9233



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-138.24(5SF)	4	CRV	10174
2	50	0	Inf. Dilution	dG:-139.90(5SF)	4	CRV	10174
3	75	0	Inf. Dilution	dG:-141.77(5SF)	4	CRV	10174

4	100	0	Inf. Dilution	dG:-143.82 (5SF)	4	CRV	10174
5	125	0	Inf. Dilution	dG:-146.03 (5SF)	4	CRV	10174
6	150	0	Inf. Dilution	dG:-148.36 (5SF)	4	CRV	10174
7	175	0	Inf. Dilution	dG:-150.81 (5SF)	4	CRV	10174
8	200	0	Inf. Dilution	dG:-153.36 (5SF)	4	CRV	10174
9	225	0	Inf. Dilution	dG:-155.99 (5SF)	4	CRV	10174
10	250	0	Inf. Dilution	dG:-158.68 (5SF)	4	CRV	10174
11	300	0	Inf. Dilution	dG:-164.12 (5SF)	4	CRV	10174

Reaction No. 9234							
2<C(s)> + 1.5<H2(g)> + Au(s) + O2(g) = Au+1_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-49.87 (4SF)	4	CRV	10174
2	50	0	Inf. Dilution	dG:-51.13 (4SF)	4	CRV	10174
3	75	0	Inf. Dilution	dG:-52.47 (4SF)	4	CRV	10174
4	100	0	Inf. Dilution	dG:-53.90 (4SF)	4	CRV	10174
5	125	0	Inf. Dilution	dG:-55.40 (4SF)	4	CRV	10174
6	150	0	Inf. Dilution	dG:-56.97 (4SF)	4	CRV	10174
7	175	0	Inf. Dilution	dG:-58.59 (4SF)	4	CRV	10174
8	200	0	Inf. Dilution	dG:-60.26 (4SF)	4	CRV	10174
9	225	0	Inf. Dilution	dG:-61.99 (4SF)	4	CRV	10174
10	250	0	Inf. Dilution	dG:-63.76 (4SF)	4	CRV	10174
11	300	0	Inf. Dilution	dG:-67.42 (4SF)	4	CRV	10174

Reaction No. 18528							
Gly-1 + Au+3 = Au+3_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.05 (4SF)	1	EOR	
2	25	0	Cl-1 salt	lgK:7.05 (3SF)	4	MGL	1555

Reaction No. 18529							
2<Gly-1> + Au+3 = Au+3_Gly-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.57 (4SF)	1	EOR	
2	25	0	Cl-1 salt	lgK:10.57 (4SF)	4	EIU	1555

Reaction No. 29082							
--------------------	--	--	--	--	--	--	--

Pyridine + Au+1 + I-1 = Au+1_Pyridine_I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Et4NC1O4	lgK:6.26 (3SF)	5	ENS	1802
2	25	0.1	Et4NC1O4	dH:7.8 (2SF)	5	ENS	1802

Reaction No. 29180							
Ala-1 + Au+3 = Au+3_Ala-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.92 (3SF)	3	MGL	2392

Reaction No. 29181							
2<Ala-1> + Au+3 = Au+3_Ala-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:9.96 (3SF)	3	MGL	2392

Reaction No. 31076							
Thiourea + Au+1 = Au+1_Thiourea							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:4.52 (3SF)	4	EOD	22560
Note (1): Zhang & Yang, Acta Chim. Sinica, 1980, 38, 521							
2	25	0.5	Unknown	lgK:4.52 (3SF)	3	CRV	2556
3	25	0.5	Unknown	dH:-30. (2SF)	3	CRV	2556

Reaction No. 36000							
2<Thiourea> + Au+1 = Au+1_Thiourea (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HCl	lgK:21.3 (3SF)	0	MSL	4119
2	23	0.45	Na2SO4	lgK:25.3 (3SF)	0	MPT	4292
3	25	0	Inf. Dilution	lgK:24.0 (3SF)	1	ELC	
4	25	0.2	HClO4	lgK:13.3 (0.1SD)	0	MPS	4113
Note (4): Something of a shambles (see pdf) but few games in town							
5	25	0.5	NaClO4	lgK:5.76 (3SF)	0	EOD	22560
Note (5): Zhang & Yang, Acta Chim. Sinica, 1980, 38, 521							
6	25	0.5	Unknown	lgK:5.76 (3SF)	0	CRV	2556
7	25	0.5	Unknown	dH:-23. (2SF)	0	CRV	2556

Reaction No. 36001							
--------------------	--	--	--	--	--	--	--

Au+1 + I-1 + Thiourea = Au+1_Thiourea_I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HCl	lgK:21.9(3SF)	3	MSL	4119

Reaction No. 37790							
Au+1_I-1(2) + Thiourea = Au+1_Thiourea_I-1 + I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	HCl	lgK:0.0(0.05SD)	0	MSL	4119
2	25	0.2	HClO4	lgK:1.93(3SF)	5	MSP	4113

Reaction No. 37791							
Au+1_I-1(2) + 2<Thiourea> = Au+1_Thiourea(2) + 2<I-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	HClO4	lgK:3.63(0.01SD)	5	MSP	4113
2	25	0.2	Unknown	lgK:3.63(3SF)	2	EOD	29531

Reaction No. 37792							
Au+1_Thiourea(2) + e-1 = Au(s) + 2<Thiourea>							
Note: Somewhat inconsistent with Au+1-Thiourea complex lgKs and deprecated by the authors themselves it is claimed. Paper is unclear as to whether it refers to Au(s) or Au(aq) but I can only suppose Au(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.5(2SF)	1	ELC	
Note (1): Corresponds to V = 0.384 V							
2	25	0	Inf. Dilution	E:0.365(3SF)	0	EFE	4113
3	25	0.2	HClO4	E:0.380(3SF)	0	MEF	4113

Reaction No. 43160							
2<Au+1> + 2<DMPS-3> = Au+1(2)_DMPS-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HNO3	lgK:45.42(4SF)	4	MPT	906
2	23	1	NH4OH	lgK:45.52(4SF)	4	MPL	906

Reaction No. 43342							
Asp-2 + Au+3 = Au+3_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.63(3SF)	5	MGL	6628

Reaction No. 43343							
Au+3_Asp-2 + Asp-2 = Au+3_Asp-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:8.60(3SF)	5	MGL	6628

Reaction No. 43344							
Au+3_Asp-2(2) + Asp-2 = Au+3_Asp-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.71(3SF)	5	MGL	6628

Reaction No. 43617							
Glu-2 + Au+3 = Au+3_Glu-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.59(3SF)	5	MGL	6628

Reaction No. 43618							
Au+3_Glu-2 + Glu-2 = Au+3_Glu-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:7.99(3SF)	5	MGL	6628

Reaction No. 43619							
Au+3_Glu-2(2) + Glu-2 = Au+3_Glu-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:6.26(3SF)	5	MGL	6628

Reaction No. 52650							
Au+1_CN-1(2) + Au+1_SCN-1(2) = Au+1(2)_SCN-1(2) + 2<CN-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.5(1SF)	1	ESC	
2	25			lgK:0.5(1SF)	0	MRX	5398
3	63			lgK:0.9(1SF)	0	MRX	5398

Reaction No. 52651							
Au+3_Cl-1(4) + 4<Br-1> = Au+3_Br-1(4) + 4<Cl-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgK:7.77(3SF)	5	MSP	5398
2	25	0.2	KCl	lgR:6.60(3SF)	5	MEF	5398

Reaction No. 52652							
Au+3_Cl-1(4) + Br-1 = Au+3_Br-1_Cl-1(3) + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	ClO4-1 salt	lgK:2.4(0.06SD)	6	MSP	5571
2	25	1	NaClO4	lgK:2.27(3SF)	4	MSP	30958
3	25	3	HClO4	lgK:2.46(3SF)	5	MSP	5398

Reaction No. 52653							
Au+3_Cl-1(4) + 2<Br-1> = Au+3_Br-1(2)_Cl-1(2) + 2<Cl-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.6(2SF)	1	ELC	
2	25	3	HClO4	lgK:4.59(3SF)	5	MSP	5398
3	20	0.1	HCl	lgR:5.8(2SF)	5	MSP	5398

Reaction No. 52654							
Au+3_Cl-1(4) + 3<Br-1> = Au+3_Br-1(3)_Cl-1 + 3<Cl-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgK:6.40(3SF)	5	MSP	5398

Reaction No. 52655							
Au+3_Cl-1(3) + Br-1 = Au+3_Br-1_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.8(2SF)	1	ELC	
2	25	0.4	HClO4	lgK:8.27(3SF)	0	MRX	5398

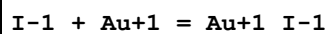
Reaction No. 52656							
Au+3_Br-1_Cl-1(2) + Br-1 = Au+3_Br-1(2)_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.4	HClO4	lgK:6.94(3SF)	5	MRX	5398

Reaction No. 52657							
Au+3_Br-1(2)_Cl-1 + Br-1 = Au+3_Br-1(3)_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.4	HClO4	lgK:6.05(3SF)	5	MRX	5398

Reaction No. 52658							
Au+3_Cl-1_CN-1(3) + I-1 = Au+3_CN-1(3)_I-1 + Cl-1							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.4	HClO ₄	lgK:4.67 (3SF)	5	MEF	6686

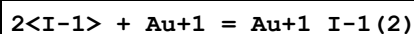
Reaction No. 52784



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	127			lgK:2.07 (3SF)	3	MPL	5398
2	20	0.1	MeCN Et ₄ NClO ₄	lgK:17.1 (3SF)	4	MEF	6758
3	25	0.1	Pyridine Et ₄ NClO ₄	lgK:6.26 (3SF)	4	MGL	22560

Note (3): Ahrland et al., Inorg. Chem., 1989, 28, 1833

Reaction No. 52785



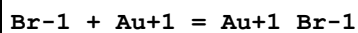
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.6 (3SF)	1	ELC	
2	127			lgK:4.23 (3SF)	3	MPL	5398
3	20	0.1	MeCN Et ₄ NClO ₄	lgK:23.8 (3SF)	4	MEF	6758
4	25	0.1	MeCN Et ₄ NClO ₄	lgK:23. (2SF)	4	MGL	22560

Note (4): Ahrland et al., Inorg. Chem., 1989, 28, 1833

5	25	0.1	Pyridine Et ₄ NClO ₄	lgK:11.9 (3SF)	4	MGL	22560
---	----	-----	---	----------------	---	-----	-------

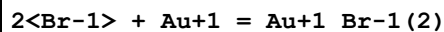
Note (5): Ahrland et al., Inorg. Chem., 1989, 28, 1833

Reaction No. 52881



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	DMSO Et ₄ NClO ₄	lgK:10.6 (3SF)	5	MIE	5398
2	25	0.1	DMSO LiClO ₄	lgK:10.6 (3SF)	5	MIE	5398
3	20	0.1	MeCN Et ₄ NClO ₄	lgK:12.0 (3SF)	5	MEF	6758

Reaction No. 52882



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

1	25	0	Inf. Dilution	lgK:20. (2SF)	0	EES	
2	25	0	Inf. Dilution	lgK:14.3 (3SF)	1	ELC	
3	25			lgK:12. (2SF)	0	EES	140
4	25	0.1	DMSO Et4NC1O4	lgK:16.6 (3SF)	5	MIE	5398
5	25	0.1	DMSO LiClO4	lgK:16.6 (3SF)	5	MIE	5398
6	20	0.1	MeCN Et4NC1O4	lgK:20.6 (3SF)	5	MEF	6758

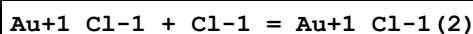
Reaction No. 52883							
4<Br-1> + Au+3 = Au+3_Br-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.81 (4SF)	1	ELC	
2	25	0.3	NaClO4	lgK:32.3 (3SF)	0	MSP	23202
Note (2): EBP: Au+3_Br-1 (4) + 3<e-1> = Au(s) + 4<Br-1>							
3	25	0.4	HClO4	lgK:27. (2SF)	0	MRX	5398
4	25			lgK:32. (2SF)	0	EES	140

Reaction No. 53056							
Cl-1 + Au+1 = Au+1_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	100	0	P=500 Inf. Dilution	lgK:6.49 (3SF)	4	EOD	6495
2	150	0	P=500 Inf. Dilution	lgK:5.93 (3SF)	4	EOD	6495
3	200	0	P=500 Inf. Dilution	lgK:5.56 (3SF)	4	EOD	6495
4	250	0	P=500 Inf. Dilution	lgK:5.34 (3SF)	3	EOD	6495
5	300	0	P=500 Inf. Dilution	lgK:5.28 (3SF)	3	EOD	6495
6	100	0	P=1000 Inf. Dilution	lgK:6.42 (3SF)	4	EOD	6495
7	150	0	P=1000 Inf. Dilution	lgK:5.85 (3SF)	4	EOD	6495
8	200	0	P=1000 Inf. Dilution	lgK:5.46 (3SF)	4	EOD	6495
9	250	0	P=1000 Inf. Dilution	lgK:5.2 (2SF)	3	EOD	6495
10	300	0	P=1000 Inf. Dilution	lgK:5.06 (3SF)	3	EOD	6495

11	100	0	P=2000 Inf. Dilution	lgK:6.33 (3SF)	4	EOD	6495
12	150	0	P=2000 Inf. Dilution	lgK:5.73 (3SF)	4	EOD	6495
13	200	0	P=2000 Inf. Dilution	lgK:5.31 (3SF)	4	EOD	6495
14	250	0	P=2000 Inf. Dilution	lgK:5.0 (2SF)	3	EOD	6495
15	300	0	P=2000 Inf. Dilution	lgK:4.8 (2SF)	3	EOD	6495
16	100	0	P=3000 Inf. Dilution	lgK:6.26 (3SF)	4	EOD	6495
17	150	0	P=3000 Inf. Dilution	lgK:5.65 (3SF)	4	EOD	6495
18	200	0	P=3000 Inf. Dilution	lgK:5.2 (2SF)	4	EOD	6495
19	250	0	P=3000 Inf. Dilution	lgK:4.87 (3SF)	3	EOD	6495
20	300	0	P=3000 Inf. Dilution	lgK:4.63 (3SF)	3	EOD	6495
21	100	0	P=4000 Inf. Dilution	lgK:6.22 (3SF)	4	EOD	6495
22	150	0	P=4000 Inf. Dilution	lgK:5.59 (3SF)	4	EOD	6495
23	200	0	P=4000 Inf. Dilution	lgK:5.13 (3SF)	4	EOD	6495
24	250	0	P=4000 Inf. Dilution	lgK:4.78 (3SF)	3	EOD	6495
25	300	0	P=4000 Inf. Dilution	lgK:4.51 (3SF)	3	EOD	6495
26	100	0	P=5000 Inf. Dilution	lgK:6.19 (3SF)	4	EOD	6495
27	150	0	P=5000 Inf. Dilution	lgK:5.55 (3SF)	4	EOD	6495
28	200	0	P=5000 Inf. Dilution	lgK:5.07 (3SF)	4	EOD	6495
29	250	0	P=5000 Inf. Dilution	lgK:4.7 (2SF)	3	EOD	6495
30	300	0	P=5000 Inf. Dilution	lgK:4.42 (3SF)	3	EOD	6495
31	0	0	Inf. Dilution	lgK:8.60 (3SF)	5	EOD	6495
32	25	0	Inf. Dilution	lgK:7.912 (4SF)	1	EOR	
33	25	0	Inf. Dilution	lgK:7.92 (3SF)	5	EOD	6495
34	50	0	Inf. Dilution	lgK:7.38 (3SF)	5	EOD	6495

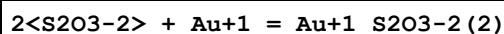
35	75	0	Inf. Dilution	lgK:6.93 (3SF)	5	EOD	6495
36	100	0	Inf. Dilution	lgK:6.57 (3SF)	4	EOD	6495
37	125	0	Inf. Dilution	lgK:6.27 (3SF)	4	EOD	6495
38	150	0	Inf. Dilution	lgK:6.03 (3SF)	4	EOD	6495
39	175	0	Inf. Dilution	lgK:5.84 (3SF)	4	EOD	6495
40	200	0	Inf. Dilution	lgK:5.69 (3SF)	4	EOD	6495
41	225	0	Inf. Dilution	lgK:5.59 (3SF)	3	EOD	6495
42	250	0	Inf. Dilution	lgK:5.54 (3SF)	3	EOD	6495
43	275	0	Inf. Dilution	lgK:5.54 (3SF)	3	EOD	6495
44	300	0	Inf. Dilution	lgK:5.62 (3SF)	3	EOD	6495
45	25	0.1	DMSO Et4NC1O4	lgK:12.6 (3SF)	5	MIE	5398
46	25	0.1	DMSO LiClO4	lgK:12.6 (3SF)	5	MIE	5398
47	20	0.1	MeCN Et4NC1O4	lgK:12.0 (3SF)	5	MEF	6758
48	25	0.1	MeCN Et4NC1O4	lgK:12.63 (4SF)	5	MRX	5398

Reaction No. 53057



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	MeCN Et4NC1O4	lgK:8.89 (3SF)	5	MRX	5398

Reaction No. 53258



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:28. (2SF)	0	EOD	20001

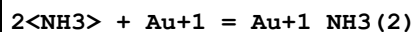
Note (2): Skibsted and Bjerrum, Acta Chem. Scand., 1977, A31, 155

3	25	0	Inf. Dilution	lgK:26.1 (3SF)	0	EOD	21129
---	----	---	---------------	----------------	---	-----	-------

Note (3): Recalculated from Acta Chem. Scand., 1977, A31, 155

4	25			lgK:26. (2SF)	3	MPL	5398
5	25:30	0.1	NaClO4	lgK:24. (2SF)	3	CRV	21011

Reaction No. 53539



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

1	22			lgK:27.(2SF)	0	EES	140
2	25	0	Inf. Dilution	lgK:14.88(4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:26.5(3SF)	0	EES	5398
4	25:30	0.1	NaClO4	lgK:26.(2SF)	0	CRV	21011
5	25:30	1	NaClO4	lgK:13.(2SF)	2	CRV	21011

Reaction No. 53540							
Au+3_NH3(3) + NH3 = Au+3_NH3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:10.3(3SF)	5	MGL	5398

Reaction No. 54308							
Au+3_Br-1(3) + Br-1 = Au+3_Br-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.47(3SF)	4	MEF	931

Reaction No. 55485							
2<C(s)> + Au(s) + N2(g) + e-1 = Au+1_CN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:68.3(0.1SD)	4	ENS	802
2	25	0	Inf. Dilution	dG:285.8(4SF) kJ	5	CRV	7421
3	25	0	Inf. Dilution	dG:285.8(4SF) kJ	4	CRV	9015
4	25	0	Inf. Dilution	dH:57.9(0.1SD)	3	ENS	802
5	25	0	Inf. Dilution	dH:242.3(4SF) kJ	4	CRV	9015

Reaction No. 57344							
0.75<Au+3_Cl-1(4)> + 0.25<Au+3_Br-1(4)> = Au+3_Br-1_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.6(0.04SD)	5	MSP	5550

Reaction No. 57345							
0.5<Au+3_Cl-1(4)> + 0.5<Au+3_Br-1(4)> = Au+3_Br-1(2)_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.0(2SF)	1	ELC	
2	25	1	NaClO4	lgK:0.7(0.05SD)	3	MSP	5550

Reaction No. 57346							
--------------------	--	--	--	--	--	--	--

0.25<Au+3_Cl-1(4)> + 0.75<Au+3_Br-1(4)> = Au+3_Br-1(3)_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:0.6(0.05SD)	5	MSP	5550

Reaction No. 57554							
AuO2(s) + 4<H+1> + 4<e-1> = Au(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:120.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:118.4(0.1SD)	0	ENS	5569

Reaction No. 57555							
Au+3_OH-1 + H+1 + 3<e-1> = Au(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:75.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:78.0(0.04SD)	0	ENS	5569
3	25	0	Inf. Dilution	E:1.44(3SF)	0	CRV	7761

Reaction No. 57556							
Au+3_OH-1(2) + 2<H+1> + 3<e-1> = Au(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:75.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:78.45(0.02SD)	0	ENS	5569

Reaction No. 57557							
Au+3_OH-1(4) + 4<H+1> + 3<e-1> = Au(s) + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:91.2(0.05SD)	0	ENS	5569

Reaction No. 57558							
Au+3_OH-1(5) + 5<H+1> + 3<e-1> = Au(s) + 5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:104.6(0.05SD)	0	ENS	5569

Reaction No. 57559							
Au+3_OH-1(6) + 6<H+1> + 3<e-1> = Au(s) + 6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:120.6(0.05SD)	0	ENS	5569

Reaction No. 57560							
$\text{Au}+1_{\text{CN}-1(2)} + 2\text{<H+1>} + \text{e-1} = \text{Au(s)} + 2\text{<H+1_{CN}-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.9 (0.05SD)	2	ENS	5569

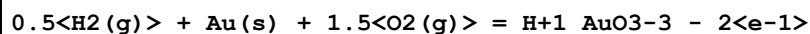
Reaction No. 57582							
$\text{Au(s)} + 0.5\text{<I2(s)>} = \text{Au}+1_{\text{I}-1}_{\text{(s)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.6 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:0.714 (3SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:0.714 (3SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:0.675 (3SF)	0	ENS	6422
5	25	0	Inf. Dilution	dG:-0.1 (0.1SD)	0	ENS	5323
6	25	0	Inf. Dilution	dG:-0.97 (2SF)	0	EFE	6422
7	25	0	Inf. Dilution	dH:0.0 (1SF) kJ	0	CRV	6330
8	25	0	Inf. Dilution	dH:0.0 (1SF)	0	MTD	6422
9	25	0	Inf. Dilution	dH:0.0 (1SF) kJ	0	CRV	9015
10	27	0	Inf. Dilution	dH:-0.46 (2SF)	0	MTD	6422
11	127	0	Inf. Dilution	dH:-1.95 (3SF)	0	MTD	6422

Reaction No. 57583							
$\text{Au(s)} + 1.5\text{<H2(g)>} + 1.5\text{<O2(g)>} = \text{Au}+3_{\text{OH}-1(3)_{\text{(s)}}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.51 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:55.05 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:36.54 (4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-75.7 (0.1SD)	4	ENS	5323
5	25	0	Inf. Dilution	dG:-75.7 (3SF)	4	EFE	6422
6	25	0	Inf. Dilution	dG:-316.92 (5SF) kJ	3	CRV	9015
7	25	0	Inf. Dilution	dH:-101.5 (4SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-424.7 (4SF) kJ	4	CRV	9015
9	127	0	Inf. Dilution	dH:-101.7 (4SF)	1	MTD	6422

Reaction No. 57584							
$\text{Au(s)} + 1.5\text{<O2(g)>} = \text{AuO3-3} - 3\text{<e-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

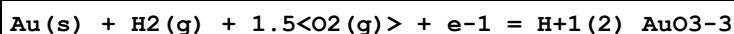
1	25	0	Inf. Dilution	dG:-12.4(0.1SD)	5	ENS	5323
2	25	0	Inf. Dilution	dG:-51.8(3SF)kJ	4	CRV	9015

Reaction No. 57585



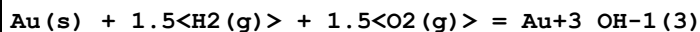
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-29.7(0.1SD)	0	ENS	5323
3	25	0	Inf. Dilution	dG:-142.2(4SF)kJ	0	CRV	9015

Reaction No. 57586



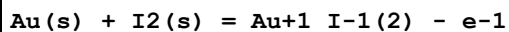
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-52.2(0.1SD)	0	ENS	5323
3	25	0	Inf. Dilution	dG:-218.3(4SF)kJ	0	CRV	9015

Reaction No. 57587



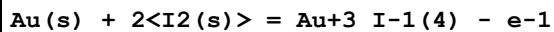
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-67.7(0.1SD)	5	ENS	5323
2	25	0	Inf. Dilution	dG:-283.37(5SF)kJ	4	CRV	9015

Reaction No. 57588



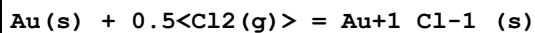
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-11.3(0.05SD)	5	ENS	5323

Reaction No. 57589



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-10.6(0.05SD)	0	ENS	5323

Reaction No. 57598



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

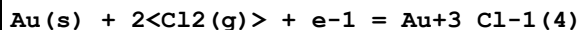
1	25	0	Inf. Dilution	lgK:2.557 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:2.518 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:0.947 (3SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:0.021 (2SF)	1	ENS	6422
5	300	0	Inf. Dilution	lgK:-0.443 (3SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-3.6 (0.05SD)	4	ENS	5323
7	25	0	Inf. Dilution	dG:-3.49 (3SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-34.7 (3SF) kJ	4	CRV	6330
9	25	0	Inf. Dilution	dH:-8.7 (2SF)	2	MTD	6422
10	25	0	Inf. Dilution	dH:-34.7 (3SF) kJ	4	CRV	9015
11	127	0	Inf. Dilution	dH:-8.5 (2SF)	1	MTD	6422
12	227	0	Inf. Dilution	dH:-8.4 (2SF)	0	MTD	6422
13	300	0	Inf. Dilution	dH:-8.3 (2SF)	0	MTD	6422

Reaction No. 57599							
Au(s) + 1.5<Cl ₂ (g)> = Au+3_Cl-1(3)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.377 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:8.250 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:3.169 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:0.171 (3SF)	2	ENS	6422
5	25	0	Inf. Dilution	dG:-10.8 (0.05SD)	3	ENS	5323
6	25	0	Inf. Dilution	dG:-11.43 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-28.3 (3SF)	2	CRV	
Note (7): [63WiB] Table 2							
8	25	0	Inf. Dilution	dH:-117.6 (4SF) kJ	4	CRV	6330
9	25	0	Inf. Dilution	dH:-28.1 (3SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-117.6 (4SF) kJ	4	CRV	9015
11	127	0	Inf. Dilution	dH:-27.7 (3SF)	2	MTD	6422
12	227	0	Inf. Dilution	dH:-27.2 (3SF)	0	MTD	6422

Reaction No. 57600							
Au(s) + Cl ₂ (g) + e-1 = Au+1_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-36.1 (0.05SD)	0	ENS	5323

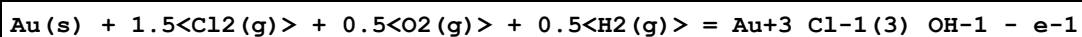
3	25	0	GAMSPATH	dG: -150.8 (4SF) kJ	0	CRV	12638
4	25	0	Inf. Dilution	dH: -151.12 (5SF) kJ	0	CRV	9015
5	25	0	GAMSPATH	dH: -182.220 (6SF) kJ	0	CRV	12638

Reaction No. 57601



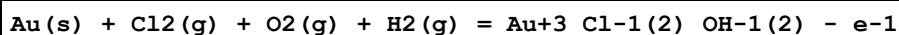
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG: -56.2 (0.05SD)	5	ENS	5323
2	25	0	Inf. Dilution	dG: -235.1 (4SF) kJ	5	CRV	7421
3	25	0	GAMSPATH	dG: -240.2 (4SF) kJ	1	CRV	12638
4	25	0	GAMSPATH	dH: -330.310 (6SF) kJ	3	CRV	12638

Reaction No. 57602



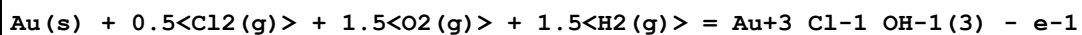
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 53.8 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG: -73.1 (0.1SD)	0	ENS	5323

Reaction No. 57603



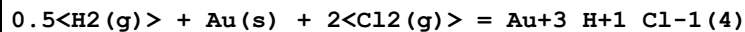
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 65.3 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG: -88.7 (0.05SD)	0	ENS	5323

Reaction No. 57604



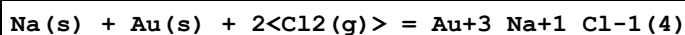
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK: 77.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG: -103.2 (0.05SD)	0	ENS	5323

Reaction No. 57605



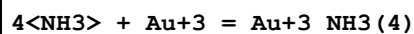
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG: -56.2 (0.05SD)	5	ENS	5323

Reaction No. 57606



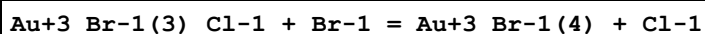
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-118.8(0.05SD)	5	ENS	5323

Reaction No. 57609



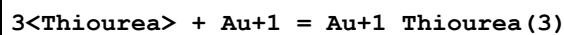
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.(2SF)	1	EES	
2	25	0	Inf. Dilution	lgK:48.0(4SF)	1	EOR	
3	25	1	(Na)ClO ₄	lgK:46.(2SF)	4	MSP	5570

Reaction No. 57610



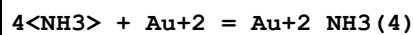
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	ClO ₄ -1 salt	lgK:1.2(0.1SD)	5	MSP	5571
2	25	1	NaClO ₄	lgK:0.95(2SF)	4	MSP	30958

Reaction No. 58322



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO ₄	lgK:6.10(3SF)	4	EOD	22560
Note (1): Zhang & Yang, Acta Chim. Sinica, 1980, 38, 521							
2	25	0.5	Unknown	lgK:6.10(3SF)	3	CRV	2556
3	25	0.5	Unknown	dH:-30.(2SF)	3	CRV	2556

Reaction No. 58324



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:30.(2SF)	3	EES	140

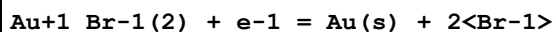
Reaction No. 58865



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.401(4SF)	0	MEF	778
3	25	0	Inf. Dilution	E:1.36(3SF)	0	CRV	3006
4	25	0	Inf. Dilution	E:1.401(4SF)	0	CRV	6330
5	25	0	Inf. Dilution	E:1.41(3SF)	0	CRV	7761

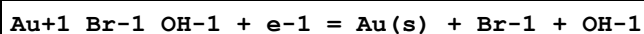
6	25	0	Inf. Dilution	E:1.36(3SF)	0	CRV	22551
---	----	---	---------------	-------------	---	-----	-------

Reaction No. 58866



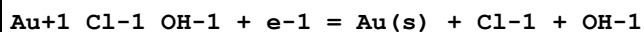
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.960(3SF)	5	CRV	3006
2	25	0	Inf. Dilution	E:0.959(3SF)	5	CRV	6330
3	25	0	Inf. Dilution	E:0.96(2SF)	4	CRV	22551
4	25	0.1:1	Br-1 salt	E:0.959(3SF)	5	MEF	778

Reaction No. 58867



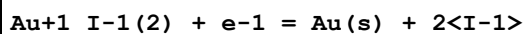
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.001:1	Br-1 salt	E:0.63(2SF)	5	MEF	778

Reaction No. 58868



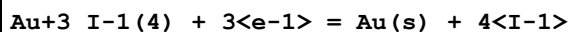
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.0096:0.79	NaCl	E:0.75(2SF)	5	MEF	778

Reaction No. 58869



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.578(3SF)	0	CRV	3006
3	25	0	Inf. Dilution	E:0.578(3SF)	0	CRV	22551
4	25	0.15:1	KI	E:0.578(3SF)	0	MEF	778

Reaction No. 58870



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.56(2SF)	0	MEF	778
3	25	0	Inf. Dilution	E:0.56(2SF)	0	CRV	3006
4	25	0	Inf. Dilution	E:0.56(2SF)	0	CRV	22551

Reaction No. 58871

Au2O3(alpha,s) + 6<H+1> + 6<e-1> = 2<Au(s)> + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:138.3(4SF)	1	EOR	
2	25	0	Inf. Dilution	E:1.35(3SF)	0	CRV	7761
3	25	2	H2SO4	E:1.360(4SF)	3	MEF	778
4	25	0	Inf. Dilution	dH:-844.9(4SF)kJ	2	CRV	7761

Reaction No. 58872							
AuO2(s) + 4<H+1> + e-1 = Au+3 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:2.507(4SF)	0	MEF	778

Reaction No. 58873							
Au+1_OH-1(2) + e-1 = Au(s) + 2<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.4(1SF)	0	MEF	778
3	25	0	Inf. Dilution	E:1.0(2SF)	0	CRV	7761

Reaction No. 58874							
Au+3_OH-1(3)_(s) + 3<H+1> + 2<e-1> = Au+1 + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.502(4SF)	0	MEF	778

Reaction No. 58875							
Au+1_SCN-1(2) + e-1 = Au(s) + 2<SCN-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.661(3SF)	5	MEF	778
2	25	0	Inf. Dilution	E:0.662(3SF)	5	CRV	3006
3	25	0	Inf. Dilution	E:0.662(3SF)	4	CRV	22551

Reaction No. 58876							
Au+3_SCN-1(4) + 3<e-1> = Au(s) + 4<SCN-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.636(3SF)	5	CRV	3006
2	25	0	Inf. Dilution	E:0.636(3SF)	4	CRV	22551

3	25	0.05:3	Cl-1 salt	E:0.636(3SF)	5	MEF	778
---	----	--------	-----------	--------------	---	-----	-----

Reaction No. 58877							
$\text{H+1_AuO3-3} + 5\text{<H+1>} + 3\text{<e-1>} = \text{Au(s)} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:106.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:2.059(4SF)	0	MEF	778

Reaction No. 58878							
$\text{H+1(2)_AuO3-3} + 4\text{<H+1>} + 3\text{<e-1>} = \text{Au(s)} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:92.9(3SF)	1	ESC	
2	25	0	Inf. Dilution	E:1.796(4SF)	0	MEF	778

Reaction No. 58879							
$\text{H+1_AuO3-3} + 5\text{<H+1>} + 2\text{<e-1>} = \text{Au+1} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.243(4SF)	5	MEF	778

Reaction No. 58880							
$\text{H+1(2)_AuO3-3} + 4\text{<H+1>} + 2\text{<e-1>} = \text{Au+1} + 3\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:62.5(3SF)	1	EDH	
2	25	0	Inf. Dilution	E:1.849(4SF)	5	MEF	778

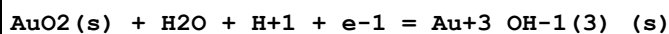
Reaction No. 58881							
$\text{Au+3_Br-1(4)} + 2\text{<e-1>} = \text{Au+1_Br-1(s)} + 3\text{<Br-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.820(3SF)	0	MEF	778

Reaction No. 58882							
$2\text{<AuO2(s)>} + 2\text{<H+1>} + 2\text{<e-1>} = \text{Au2O3(alpha,s)} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.630(4SF)	0	MEF	778

Reaction No. 58883							
$2\text{<AuO2(s)>} + 2\text{<H+1>} + 2\text{<e-1>} = \text{Au2O3(beta,s)} + \text{H2O}$							

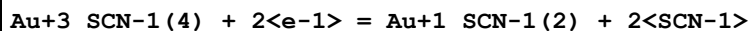
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.465 (4SF)	5	MEF	778

Reaction No. 58884



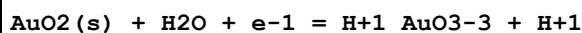
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:2.305 (4SF)	0	MEF	778

Reaction No. 58885



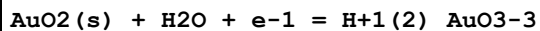
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.623 (3SF)	2	CRV	3006
2	25	0	Inf. Dilution	E:-0.623 (3SF)	0	CRV	22551
Note (2): No background electrolyte; wrong sign (see analog. Br ⁻ & I ⁻ rxns)							
3	25	0.3:3	NaSCN	E:0.623 (3SF)	5	MEF	778

Reaction No. 58886



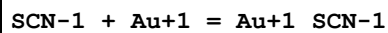
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.822 (3SF)	5	MEF	778

Reaction No. 58887



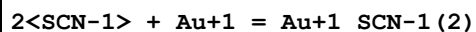
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.611 (4SF)	5	MEF	778

Reaction No. 58917



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.43 (4SF)	1	EOR	
2	25	3	Unknown	lgK:15.27 (4SF)	4	CRV	552

Reaction No. 58918



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.2 (3SF)	1	ELC	
2	25	3	Unknown	lgK:16.98 (4SF)	4	CRV	552

Reaction No. 58985							
Au+3_NH3(4) + H2O = Au+3_NH3(3)_OH-1 + H+1_NH3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50:70	1	NaClO4	lgK:0.3(1SF)	4	MPS	5398
2	80	1	NaClO4	lgK:0.0(1SF)	4	MPS	5398

Reaction No. 59000							
Au+1_Br-1(2) + OH-1 = Au+1_Br-1_OH-1 + Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:5.53(3SF)	0	MEF	5398

Reaction No. 59001							
Au+1_Cl-1(2) + OH-1 = Au+1_Cl-1_OH-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.66(3SF)	0	MEF	5398

Reaction No. 59002							
Au+1_Cl-1_OH-1 + OH-1 = Au+1_OH-1(2) + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.(1SF)	0	MEF	5398

Reaction No. 59208							
Pen-2 + Au+1 = Au+1_Pen-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.15	KNO3	lgK:12.50(4SF)	4	MLC	3549
2	37	0.15	KNO3	lgK:13.50(4SF)	4	MLC	3549

Reaction No. 59209							
Cys-2 + Au+1 = Au+1_Cys-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.15	KNO3	lgK:11.11(4SF)	4	MLC	3549
2	37	0.15	KNO3	lgK:12.04(4SF)	4	MLC	3549

Reaction No. 59210							
2SHSuccin-3 + Au+1 = Au+1_2SHSuccin-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.15	KNO3	lgK:10.27(4SF)	4	MLC	3549

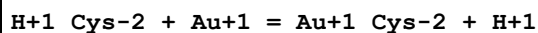
2	37	0.15	KNO3	lgK:11.23 (4SF)	4	MLC	3549
---	----	------	------	-----------------	---	-----	------

Reaction No. 59235



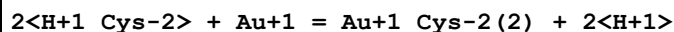
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:2.68 (3SF)	0	MSL	905

Reaction No. 59236



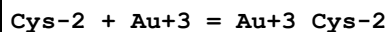
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:24.18 (4SF)	3	MMR	905

Reaction No. 59237



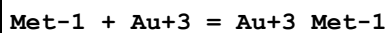
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:28.48 (4SF)	3	MMR	905

Reaction No. 59238



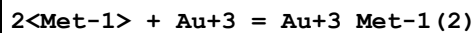
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	0.1	KNO3	lgK:14.85 (4SF)	5	MGL	905
2	30	0.1	KNO3	lgK:14.85 (4SF)	5	MGL	905

Reaction No. 59372



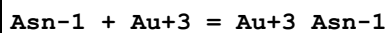
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:7.23 (3SF)	4	MGL	905

Reaction No. 59373



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:10.19 (4SF)	4	MGL	905

Reaction No. 59549



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:6.18 (3SF)	0	MGL	905
2	25	0.1	NaClO4	lgK:8.88 (3SF)	4	MGL	905

Reaction No. 59550							
2<Asn-1> + Au+3 = Au+3_Asn-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.02	Unknown	lgK:9.28(3SF)	0	MGL	905
2	25	0.1	NaClO4	lgK:17.02(4SF)	4	MGL	905

Reaction No. 59551							
3<Asn-1> + Au+3 = Au+3_Asn-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:24.62(4SF)	4	MGL	905

Reaction No. 60000							
Val-1 + Au+3 = Au+3_Val-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.80(3SF)	3	MGL	2392

Reaction No. 60001							
2<Val-1> + Au+3 = Au+3_Val-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:9.85(3SF)	3	MGL	2392

Reaction No. 60017							
Leu-1 + Au+3 = Au+3_Leu-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:7.14(3SF)	3	MGL	2392

Reaction No. 60018							
2<Leu-1> + Au+3 = Au+3_Leu-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:10.18(4SF)	3	MGL	2392

Reaction No. 60073							
bAla-1 + Au+3 = Au+3_bAla-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.85(3SF)	3	MGL	2392

Reaction No. 60074							
--------------------	--	--	--	--	--	--	--

2<bAla-1> + Au+3 = Au+3_bAla-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:10.46(4SF)	3	MGL	2392

Reaction No. 60512							
2<Au(CH3)2+1> + 2<OH-1> = Au(CH3)2+1(2)_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Unknown	lgK:18.57(4SF)	0	CRV	2556

Reaction No. 60751							
Au+3_NH3(4)_OH-1 + H+1 = Au+3_NH3(4) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:7.48(3SF)	4	CRV	2556

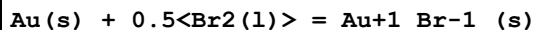
Reaction No. 65710							
Au+1(2)_SCN-1(2)_(s) = 2<Au+1> + 2<SCN-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.52(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-19.52(4SF)	4	CRV	552
3	25	1	Unknown	lgK:-19.00(4SF)	4	CRV	552

Reaction No. 66021							
2<CN-1> + Au+1 = Au+1_CN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.4(3SF)	1	ELC	
2	25	0.025	CN- salt	lgK:36.6(3SF)w	0	MPH	884

Reaction No. 66495							
3<Au(s)> + As(s) + 2<O2(g)> = Au+1(3)_AsO4-3_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.16(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:51.74(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:35.06(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:25.11(4SF)	1	ENS	6422
5	296	0	Inf. Dilution	lgK:20.32(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-71.16(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-91.8(3SF)	4	MTD	6422

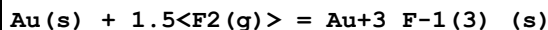
8	127	0	Inf. Dilution	dH:-91.3 (3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-90.8 (3SF)	0	MTD	6422
10	296	0	Inf. Dilution	dH:-90.2 (3SF)	0	MTD	6422

Reaction No. 66496



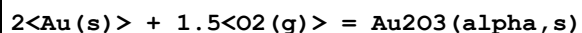
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.168 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1.153 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:0.132 (3SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-1.59 (3SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-14.0 (3SF) kJ	4	CRV	6330
6	25	0	Inf. Dilution	dH:-3.4 (2SF)	3	MTD	6422
7	25	0	Inf. Dilution	dH:-13.97 (4SF) kJ	4	CRV	9015
8	127	0	Inf. Dilution	dH:-6.9 (2SF)	0	MTD	6422

Reaction No. 66497



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.30 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:50.90 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:35.12 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:25.69 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:19.45 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-69.98 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-363.6 (4SF) kJ	4	CRV	6330
8	25	0	Inf. Dilution	dH:-86.9 (3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-363.6 (4SF) kJ	4	CRV	9015
10	127	0	Inf. Dilution	dH:-86.4 (3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-86.0 (3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-85.5 (3SF)	0	MTD	6422

Reaction No. 66499



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.64 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:-13.65 (4SF)	0	ENS	6422

3	127	0	Inf. Dilution	lgK:-13.75 (4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:-13.76 (4SF)	2	ENS	6422
5	25	0	Inf. Dilution	dG:18.61 (4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-0.80 (2SF)	4	MTD	6422
7	127	0	Inf. Dilution	dH:-0.33 (2SF)	0	MTD	6422
8	227	0	Inf. Dilution	dH:-0.14 (2SF)	0	MTD	6422

Reaction No. 66500							
Au(s) + Se(s) = AuSe(alpha,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.686 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1.671 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:1.069 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:0.697 (3SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:0.348 (3SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2.30 (3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-3.3 (2SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-3.3 (2SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-4.7 (2SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-4.8 (2SF)	4	MTD	6422

Reaction No. 66501							
Au(s) + Se(s) = AuSe(beta,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.654 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1.642 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:1.168 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:0.872 (3SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:0.574 (3SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2.26 (3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2.6 (2SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2.6 (2SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-4.0 (2SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-4.1 (2SF)	4	MTD	6422

Reaction No. 66502							
Au(s) + 2<Te(s)> = AuTe2(s)							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.012 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:2.992 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:2.180 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:1.688 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:1.354 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-4.11 (3SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-4.5 (2SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-4.5 (2SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-4.5 (2SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-4.6 (2SF)	4	MTD	6422

Reaction No. 67310							
Au(s) + 0.25<O2(g)> + H+1 = Au+1 + 0.5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.6 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.81 (3SF)	0	EFE	5
3	50	0	Inf. Dilution	lgK:-7.08 (3SF)	0	EFE	5
4	60	0	Inf. Dilution	lgK:-6.82 (3SF)	0	EFE	5
5	100	0	Inf. Dilution	lgK:-5.89 (3SF)	0	EFE	5
6	150	0	Inf. Dilution	lgK:-4.94 (3SF)	0	EFE	5
7	200	0	Inf. Dilution	lgK:-4.17 (3SF)	0	EFE	5
8	250	0	Inf. Dilution	lgK:-3.52 (3SF)	0	EFE	5
9	300	0	Inf. Dilution	lgK:-2.97 (3SF)	0	EFE	5
10	25	0	Inf. Dilution	dH:12.680 (5SF)	2	EFE	5

Reaction No. 67776							
3<Cl-1> + Au+1 = Au+1_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	100	0	P=500 Inf. Dilution	lgK:7.18 (3SF)	5	EOD	6495
2	150	0	P=500 Inf. Dilution	lgK:6.5 (2SF)	5	EOD	6495
3	200	0	P=500 Inf. Dilution	lgK:6.1 (2SF)	5	EOD	6495
4	250	0	P=500 Inf. Dilution	lgK:5.91 (3SF)	3	EOD	6495
5	300	0	P=500 Inf. Dilution	lgK:5.92 (3SF)	3	EOD	6495

6	100	0	P=1000 Inf. Dilution	lgK: 6.94 (3SF)	4	EOD	6495
7	150	0	P=1000 Inf. Dilution	lgK: 6.26 (3SF)	4	EOD	6495
8	200	0	P=1000 Inf. Dilution	lgK: 5.84 (3SF)	4	EOD	6495
9	250	0	P=1000 Inf. Dilution	lgK: 5.61 (3SF)	3	EOD	6495
10	300	0	P=1000 Inf. Dilution	lgK: 5.53 (3SF)	3	EOD	6495
11	100	0	P=2000 Inf. Dilution	lgK: 6.56 (3SF)	4	EOD	6495
12	150	0	P=2000 Inf. Dilution	lgK: 5.88 (3SF)	4	EOD	6495
13	200	0	P=2000 Inf. Dilution	lgK: 5.43 (3SF)	4	EOD	6495
14	250	0	P=2000 Inf. Dilution	lgK: 5.16 (3SF)	3	EOD	6495
15	300	0	P=2000 Inf. Dilution	lgK: 5.02 (3SF)	3	EOD	6495
16	100	0	P=3000 Inf. Dilution	lgK: 6.27 (3SF)	4	EOD	6495
17	150	0	P=3000 Inf. Dilution	lgK: 5.58 (3SF)	4	EOD	6495
18	200	0	P=3000 Inf. Dilution	lgK: 5.12 (3SF)	4	EOD	6495
19	250	0	P=3000 Inf. Dilution	lgK: 4.82 (3SF)	3	EOD	6495
20	300	0	P=3000 Inf. Dilution	lgK: 4.64 (3SF)	3	EOD	6495
21	100	0	P=4000 Inf. Dilution	lgK: 6.03 (3SF)	4	EOD	6495
22	150	0	P=4000 Inf. Dilution	lgK: 5.33 (3SF)	4	EOD	6495
23	200	0	P=4000 Inf. Dilution	lgK: 4.86 (3SF)	4	EOD	6495
24	250	0	P=4000 Inf. Dilution	lgK: 4.55 (3SF)	3	EOD	6495
25	300	0	P=4000 Inf. Dilution	lgK: 4.35 (3SF)	3	EOD	6495
26	100	0	P=5000 Inf. Dilution	lgK: 5.84 (3SF)	4	EOD	6495
27	150	0	P=5000 Inf. Dilution	lgK: 5.12 (3SF)	4	EOD	6495
28	200	0	P=5000 Inf. Dilution	lgK: 4.65 (3SF)	4	EOD	6495

29	250	0	P=5000 Inf. Dilution	lgK:4.32 (3SF)	3	EOD	6495
30	300	0	P=5000 Inf. Dilution	lgK:4.1 (2SF)	3	EOD	6495
31	0	0	Inf. Dilution	lgK:10.3 (3SF)	5	EOD	6495
32	25	0	Inf. Dilution	lgK:9.298 (4SF)	1	EOR	
33	25	0	Inf. Dilution	lgK:9.29 (3SF)	5	EOD	6495
34	50	0	Inf. Dilution	lgK:8.54 (3SF)	5	EOD	6495
35	75	0	Inf. Dilution	lgK:7.94 (3SF)	5	EOD	6495
36	100	0	Inf. Dilution	lgK:7.46 (3SF)	4	EOD	6495
37	125	0	Inf. Dilution	lgK:7.08 (3SF)	4	EOD	6495
38	150	0	Inf. Dilution	lgK:6.79 (3SF)	4	EOD	6495
39	175	0	Inf. Dilution	lgK:6.57 (3SF)	4	EOD	6495
40	200	0	Inf. Dilution	lgK:6.41 (3SF)	4	EOD	6495
41	225	0	Inf. Dilution	lgK:6.32 (3SF)	3	EOD	6495
42	250	0	Inf. Dilution	lgK:6.29 (3SF)	3	EOD	6495
43	275	0	Inf. Dilution	lgK:6.33 (3SF)	3	EOD	6495
44	300	0	Inf. Dilution	lgK:6.45 (3SF)	3	EOD	6495

Reaction No. 67777							
4<Cl-1> + Au+1 = Au+1_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	100	0	P=500 Inf. Dilution	lgK:-33.8 (3SF)	4	EOD	6495
2	150	0	P=500 Inf. Dilution	lgK:-29.7 (3SF)	4	EOD	6495
3	200	0	P=500 Inf. Dilution	lgK:-26.0 (3SF)	4	EOD	6495
4	250	0	P=500 Inf. Dilution	lgK:-22.5 (3SF)	3	EOD	6495
5	100	0	P=1000 Inf. Dilution	lgK:-33.5 (3SF)	4	EOD	6495
6	150	0	P=1000 Inf. Dilution	lgK:-29.4 (3SF)	4	EOD	6495
7	200	0	P=1000 Inf. Dilution	lgK:-25.9 (3SF)	4	EOD	6495
8	250	0	P=1000 Inf. Dilution	lgK:-22.7 (3SF)	3	EOD	6495
9	100	0	P=2000 Inf. Dilution	lgK:-32.8 (3SF)	4	EOD	6495

10	150	0	P=2000 Inf. Dilution	lgK:-28.9 (3SF)	4	EOD	6495
11	200	0	P=2000 Inf. Dilution	lgK:-25.6 (3SF)	4	EOD	6495
12	250	0	P=2000 Inf. Dilution	lgK:-22.6 (3SF)	3	EOD	6495
13	100	0	P=3000 Inf. Dilution	lgK:-32. (2SF)	4	EOD	6495
14	150	0	P=3000 Inf. Dilution	lgK:-28.4 (3SF)	4	EOD	6495
15	200	0	P=3000 Inf. Dilution	lgK:-25.2 (3SF)	4	EOD	6495
16	250	0	P=3000 Inf. Dilution	lgK:-22.4 (3SF)	3	EOD	6495
17	100	0	P=4000 Inf. Dilution	lgK:-31.3 (3SF)	4	EOD	6495
18	150	0	P=4000 Inf. Dilution	lgK:-27.8 (3SF)	4	EOD	6495
19	200	0	P=4000 Inf. Dilution	lgK:-24.7 (3SF)	4	EOD	6495
20	250	0	P=4000 Inf. Dilution	lgK:-22.1 (3SF)	3	EOD	6495
21	300	0	P=4000 Inf. Dilution	lgK:-19.8 (3SF)	3	EOD	6495
22	100	0	P=5000 Inf. Dilution	lgK:-30.5 (3SF)	4	EOD	6495
23	150	0	P=5000 Inf. Dilution	lgK:-27.1 (3SF)	4	EOD	6495
24	200	0	P=5000 Inf. Dilution	lgK:-24.2 (3SF)	4	EOD	6495
25	250	0	P=5000 Inf. Dilution	lgK:-21.7 (3SF)	3	EOD	6495
26	0	0	Inf. Dilution	lgK:-45.1 (3SF)	5	EOD	6495
27	25	0	Inf. Dilution	lgK:-42.0 (4SF)	1	EOR	
28	25	0	Inf. Dilution	lgK:-42. (2SF)	5	EOD	6495
29	50	0	Inf. Dilution	lgK:-39.1 (3SF)	5	EOD	6495
30	75	0	Inf. Dilution	lgK:-36.5 (3SF)	5	EOD	6495
31	100	0	Inf. Dilution	lgK:-34.1 (3SF)	4	EOD	6495
32	125	0	Inf. Dilution	lgK:-31.9 (3SF)	4	EOD	6495
33	150	0	Inf. Dilution	lgK:-29.8 (3SF)	3	EOD	6495
34	175	0	Inf. Dilution	lgK:-27.8 (3SF)	2	EOD	6495
35	200	0	Inf. Dilution	lgK:-25.9 (3SF)	2	EOD	6495
36	225	0	Inf. Dilution	lgK:-24. (2SF)	1	EOD	6495

37	250	0	Inf. Dilution	lgK:-22.2 (3SF)	1	EOD	6495
38	275	0	Inf. Dilution	lgK:-20.2 (3SF)	0	EOD	6495
39	300	0	Inf. Dilution	lgK:-18.2 (3SF)	0	EOD	6495

Reaction No. 68858							
Au+3_Cl-1_CN-1(3) = Au+3_CN-1(3) + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.4	HClO4	lgK:-5.43 (3SF)	5	MEF	6686

Reaction No. 68859							
Au+3_CN-1(3)_I-1 = Au+3_CN-1(3) + I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.4	HClO4	lgK:-10.11 (4SF)	5	MEF	6686

Reaction No. 68871							
Au+3_Dien_Cl-1 + Br-1 = Au+3_Dien_Br-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Methanol	lgK:1.49 (3SF)	4	MKN	6684

Reaction No. 68872							
Au+3_Dien_Cl-1 + SCN-1 = Au+3_Dien_SCN-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Methanol	lgK:1.98 (3SF)	4	MKN	6684

Reaction No. 68873							
Au+3_Dien_Cl-1 + N3-1 = Au+3_Dien_N3-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25		Methanol	lgK:0.87 (2SF)	4	MKN	6684

Reaction No. 70262							
Au+3_I-1(4) = Au+1_I-1(2) + I2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.5 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.3 (2SF)	0	ENS	7276

Reaction No. 70282							
Au(s) = Au(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dG:326.3(4SF) kJ	3	CRV	6330
2	25	0	Inf. Dilution	dG:326.3(4SF) kJ	4	EFE	7275
3	25	0	Inf. Dilution	dG:326.3(4SF) kJ	4	CRV	9015
4	25	0	Inf. Dilution	dH:366.1(4SF) kJ	3	CRV	6330
5	25	0	Inf. Dilution	dH:366.1(4SF) kJ	4	EFE	7275
6	25	0	Inf. Dilution	dH:366.1(4SF) kJ	4	CRV	9015

Reaction No. 70715							
Au(s) + 1.5<Br2(l)> = Au+3_Br-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-53.3(3SF) kJ	4	CRV	6330
2	25	0	Inf. Dilution	dH:-53.26(4SF) kJ	4	CRV	9015

Reaction No. 71188							
Au+3_OH-1 + H+1 + 2<e-1> = Au+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.32(3SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:1.32(3SF)	5	CRV	7761

Reaction No. 71189							
Au+2 + e-1 = Au+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.8(2SF)	4	CRV	6330
2	25	0	Inf. Dilution	E:1.8(2SF)	4	CRV	7761

Reaction No. 71229							
Au(s) + e-1 = Au-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.4(2SF)	3	CRV	7761

Reaction No. 71312							
Au2O3(am.,s) + 6<H+1> + 6<e-1> = 2<Au(s)> + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.363(4SF)	5	CRV	7761
2	25	0	Inf. Dilution	dH:-865.5(4SF) kJ	5	CRV	7761

Reaction No. 71313							
--------------------	--	--	--	--	--	--	--

Au2O3(am.,s) + 6<H+1> + 4<e-1> = 2<Au+1> + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.20(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:-474.6(4SF)kJ	2	CRV	7761

Reaction No. 71314							
Au2O3(alpha,s) + 6<H+1> + 4<e-1> = 2<Au+1> + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.18(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:-455.4(4SF)kJ	2	CRV	7761

Reaction No. 71315							
Au+3 + e-1 = Au+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.0(2SF)	0	CRV	7761

Reaction No. 71316							
Au+3_OH-1(4) + 3<e-1> = Au(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.600(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:-277.2(4SF)kJ	2	CRV	7761

Reaction No. 71317							
Au2O3(am.,s) + 3<H2O> + 6<e-1> = 2<Au(s)> + 6<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.535(3SF)	5	CRV	7761
2	25	0	Inf. Dilution	dH:-530.5(4SF)kJ	5	CRV	7761

Reaction No. 71318							
Au2O3(alpha,s) + 3<H2O> + 6<e-1> = 2<Au(s)> + 6<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:54.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.52(2SF)	0	CRV	7761

3	25	0	Inf. Dilution	dH:-508.7(4SF)kJ	2	CRV	7761
---	----	---	---------------	------------------	---	-----	------

Reaction No. 71319							
Au+3_OH-1(4) + 2<e-1> = Au+1_OH-1(2) + 2<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.4(1SF)	0	CRV	7761

Reaction No. 72263							
Au+3 + H2O = Au+3_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:7.0(2SF)	3	CRV	6478

Reaction No. 72278							
Au+3_OH-1(3)_(s) + 3<H+1> = Au+3 + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.51(0.07SD)	0	CRV	6478

Reaction No. 72374							
Au(s) + H+1 + 2<Cl-1> + 0.25<O2(g)> = Au+1_Cl-1(2) + 0.5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.07(3SF)	0	EIE	9031
3	50	0	Inf. Dilution	lgK:1.29(3SF)	0	EIE	9031
4	100	0	Inf. Dilution	lgK:1.62(3SF)	2	EIE	9031
5	150	0	Inf. Dilution	lgK:2.01(3SF)	0	EIE	9031
6	200	0	Inf. Dilution	lgK:2.53(3SF)	0	EIE	9031

Reaction No. 72375							
Au(s) + 3<H+1> + 4<Cl-1> + 0.75<O2(g)> = Au+3_Cl-1(4) + 1.5<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:10.61(4SF)	0	EIE	9031
3	50	0	Inf. Dilution	lgK:10.20(4SF)	0	EIE	9031
4	100	0	Inf. Dilution	lgK:9.31(3SF)	3	EIE	9031
5	150	0	Inf. Dilution	lgK:8.88(3SF)	0	EIE	9031

6	200	0	Inf. Dilution	lgK:9.08 (3SF)	0	EIE	9031
---	-----	---	---------------	----------------	---	-----	------

Reaction No. 74131							
Au+3_NH3(4) + 3<e-1> = Au(s) + 4<NH3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.325 (3SF)	0	CRV	22551

Reaction No. 74132							
Au+3_I-1(4) + 2<e-1> = Au+1_I-1(2) + 2<I-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.6 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.55 (2SF)	0	CRV	3006
3	25	0	Inf. Dilution	E:0.55 (2SF)	0	CRV	22551

Reaction No. 74133							
Au+1_NH3(2) + e-1 = Au(s) + 2<NH3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	10	NH4NO3	E:0.563 (3SF)	4	CRV	22551

Reaction No. 74134							
Au+1_Cl-1_(s) = Au+1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.7 (2SF)	0	CRV	22551

Reaction No. 74135							
Au+1_Br-1_(s) = Au+1 + Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-16.3 (3SF)	0	CRV	22551

Reaction No. 74136							
Au+3_Cl-1(3)_(s) = Au+3 + 3<Cl-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.3 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-24.5 (3SF)	0	CRV	22551

Reaction No. 74137							
$\text{Au}^{+3}\text{Br}^{-1}(3)_{(s)} = \text{Au}^{+3} + 3\text{Br}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -22.8(4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: -22.8(3\text{SF})$	4	CRV	22551

Reaction No. 74138							
$\text{Au}^{+3}\text{I}^{-1}(3)_{(s)} = \text{Au}^{+3} + 3\text{I}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -46.0(4\text{SF})$	1	EOR	
2	25	0	Inf. Dilution	$\lg K: -46.(2\text{SF})$	4	CRV	22551

Reaction No. 74281							
$\text{Au}^{+1}\text{I}^{-1}_{(s)} + \text{e}^{-1} = \text{Au}_{(s)} + \text{I}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 10.7(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: 0.530(3\text{SF})$	0	CRV	3006

Reaction No. 75259							
$3\text{Cl}^{-1} + \text{Br}^{-1} + \text{Au}^{+3} = \text{Au}^{+3}\text{Br}^{-1}\text{Cl}^{-1}(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 27.7(3\text{SF})$	1	ELC	
2	25	0.3	NaClO4	$\lg K: 27.43(0.1\text{SD})$	0	MSP	23202

Reaction No. 75260							
$2\text{Cl}^{-1} + 2\text{Br}^{-1} + \text{Au}^{+3} = \text{Au}^{+3}\text{Br}^{-1}(2)\text{Cl}^{-1}(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 31.11(4\text{SF})$	1	EOR	
2	25	0.3	NaClO4	$\lg K: 29.46(4\text{SF})$	7	MSP	23202

Reaction No. 75261							
$\text{Cl}^{-1} + 3\text{Br}^{-1} + \text{Au}^{+3} = \text{Au}^{+3}\text{Br}^{-1}(3)\text{Cl}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 31.8(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: 32.68(4\text{SF})$	0	EOR	
3	25	0.3	NaClO4	$\lg K: 31.03(4\text{SF})$	0	MSP	23202

Reaction No. 75262							
$3\text{<Cl-1>} + \text{Au+3} + \text{OH-1} = \text{Au+3_Cl-1(3)_OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.4 (3SF)	1	ELC	
2	25	0.3	NaClO4	lgK:33.34 (4SF)	0	MSP	23202

Reaction No. 75263							
$2\text{<Cl-1>} + \text{Au+3} + 2\text{<OH-1>} = \text{Au+3_Cl-1(2)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:40.3 (3SF)	1	ELC	
2	25	0.3	NaClO4	lgK:40.30 (4SF)	0	MSP	23202

Reaction No. 75264							
$3\text{<OH-1>} + \text{Cl-1} + \text{Au+3} = \text{Au+3_Cl-1_OH-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.6 (4SF)	1	ELC	
2	25	0.3	NaClO4	lgK:45.94 (4SF)	0	MSP	23202

Reaction No. 75265							
$4\text{<OH-1>} + \text{Au+3} = \text{Au+3_OH-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.0 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:53.0 (3SF)	1	EOR	
3	25	0.3	NaClO4	lgK:51.35 (4SF)	3	MSP	23202

Reaction No. 75266							
$\text{OH-1} + 3\text{<Br-1>} + \text{Au+3} = \text{Au+3_Br-1(3)_OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.25 (4SF)	1	EOR	
2	25	0.3	NaClO4	lgK:39.6 (3SF)	7	MSP	23202

Reaction No. 75267							
$2\text{<OH-1>} + 2\text{<Br-1>} + \text{Au+3} = \text{Au+3_Br-1(2)_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.45 (4SF)	1	EOR	
2	25	0.3	NaClO4	lgK:45.8 (3SF)	7	MSP	23202

Reaction No. 75268							
$3\text{<OH-1>} + \text{Br-1} + \text{Au+3} = \text{Au+3_Br-1_OH-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:52.55 (4SF)	1	EOR	
2	25	0.3	NaClO4	lgK:50.9 (3SF)	7	MSP	23202

Reaction No. 75797							
$\text{Au+3} + 2\text{<Glu-2>} = \text{Au+3_Glu-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:17.58 (4SF)	5	MGL	6628

Reaction No. 75826							
$\text{Au+3} + 2\text{<Asp-2>} = \text{Au+3_Asp-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:18.23 (4SF)	5	MGL	5763

Reaction No. 75959							
$2\text{<OH-1>} + \text{Au+1} = \text{Au+1_OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.0 (4SF)	1	EOR	
2	25:30	0	Inf. Dilution	lgK:22. (2SF)	3	CRV	21011

Reaction No. 75960							
$\text{OH-1} + \text{Au+1} = \text{Au+1_OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.6 (3SF)	1	EES	
2	25	0	Inf. Dilution	lgK:19.6 (4SF)	1	ELC	
3	25:30	0	Inf. Dilution	lgK:10.2 (3SF)	0	CRV	21011
4	25:30	0	Inf. Dilution	lgK:20.6 (3SF)	0	CRV	21011

Reaction No. 75961							
$\text{Au(s)} + \text{H}_2\text{O} = \text{Au+1_OH-1} + 0.5\text{<H}_2\text{(g)>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	P=500 Inf. Dilution	lgK:-22. (1.SD)	0	EVK	18069
2	50	0	P=500 Inf. Dilution	lgK:-19.5 (1.SD)	0	EVK	18069

3	100	0	P=500 Inf. Dilution	lgK: -15.6 (1.SD)	0	EVK	18069
4	150	0	P=500 Inf. Dilution	lgK: -12.9 (1.SD)	0	EVK	18069
5	200	0	P=500 Inf. Dilution	lgK: -11.0 (2SF)	0	EVK	18069
6	250	0	P=500 Inf. Dilution	lgK: -9.7 (2SF)	0	EVK	18069
7	300	0	P=500 Inf. Dilution	lgK: -8.73 (3SF)	0	EVK	18069
8	350	0	P=500 Inf. Dilution	lgK: -8.10 (3SF)	0	EVK	18069
9	400	0	P=500 Inf. Dilution	lgK: -7.71 (3SF)	0	EVK	18069
10	450	0	P=500 Inf. Dilution	lgK: -7.53 (3SF)	0	EVK	18069
11	500	0	P=500 Inf. Dilution	lgK: -7.49 (3SF)	0	EVK	18069
12	25	0	P=1000 Inf. Dilution	lgK: -22.0 (1.SD)	0	EVK	18069
13	50	0	P=1000 Inf. Dilution	lgK: -19.5 (1.SD)	0	EVK	18069
14	100	0	P=1000 Inf. Dilution	lgK: -15.7 (1.SD)	0	EVK	18069
15	150	0	P=1000 Inf. Dilution	lgK: -13.0 (1.SD)	0	EVK	18069
16	200	0	P=1000 Inf. Dilution	lgK: -11.1 (3SF)	0	EVK	18069
17	250	0	P=1000 Inf. Dilution	lgK: -9.7 (2SF)	0	EVK	18069
18	300	0	P=1000 Inf. Dilution	lgK: -8.74 (3SF)	0	EVK	18069
19	350	0	P=1000 Inf. Dilution	lgK: -8.10 (3SF)	0	EVK	18069
20	400	0	P=1000 Inf. Dilution	lgK: -7.71 (3SF)	0	EVK	18069
21	450	0	P=1000 Inf. Dilution	lgK: -7.52 (3SF)	0	EVK	18069
22	500	0	P=1000 Inf. Dilution	lgK: -7.47 (3SF)	0	EVK	18069
23	550	0	P=1000 Inf. Dilution	lgK: -7.57 (3SF)	0	EVK	18069
24	600	0	P=1000 Inf. Dilution	lgK: -7.76 (3SF)	0	EVK	18069
25	25	0	P=1500 Inf. Dilution	lgK: -22.1 (3SF)	0	EVK	18069

26	50	0	P=1500 Inf. Dilution	lgK: -19.5 (1.SD)	0	EVK	18069
27	100	0	P=1500 Inf. Dilution	lgK: -15.7 (1.SD)	0	EVK	18069
28	150	0	P=1500 Inf. Dilution	lgK: -13.0 (1.SD)	0	EVK	18069
29	200	0	P=1500 Inf. Dilution	lgK: -11.1 (2SF)	0	EVK	18069
30	250	0	P=1500 Inf. Dilution	lgK: -9.7 (2SF)	0	EVK	18069
31	300	0	P=1500 Inf. Dilution	lgK: -8.75 (3SF)	0	EVK	18069
32	350	0	P=1500 Inf. Dilution	lgK: -8.10 (3SF)	0	EVK	18069
33	400	0	P=1500 Inf. Dilution	lgK: -7.71 (3SF)	0	EVK	18069
34	450	0	P=1500 Inf. Dilution	lgK: -7.51 (3SF)	0	EVK	18069
35	500	0	P=1500 Inf. Dilution	lgK: -7.45 (3SF)	0	EVK	18069
36	550	0	P=1500 Inf. Dilution	lgK: -7.56 (3SF)	0	EVK	18069
37	600	0	P=1500 Inf. Dilution	lgK: -7.73 (3SF)	0	EVK	18069
38	25	0	P=2000 Inf. Dilution	lgK: -22.1 (1.SD)	0	EVK	18069
39	50	0	P=2000 Inf. Dilution	lgK: -19.6 (1.SD)	0	EVK	18069
40	100	0	P=2000 Inf. Dilution	lgK: -15.7 (1.SD)	0	EVK	18069
41	150	0	P=2000 Inf. Dilution	lgK: -13.0 (1.SD)	0	EVK	18069
42	200	0	P=2000 Inf. Dilution	lgK: -11.1 (2SF)	0	EVK	18069
43	250	0	P=2000 Inf. Dilution	lgK: -9.7 (2SF)	0	EVK	18069
44	300	0	P=2000 Inf. Dilution	lgK: -8.8 (2SF)	0	EVK	18069
45	350	0	P=2000 Inf. Dilution	lgK: -8.1 (2SF)	0	EVK	18069
46	400	0	P=2000 Inf. Dilution	lgK: -7.7 (2SF)	0	EVK	18069
47	450	0	P=2000 Inf. Dilution	lgK: -7.5 (2SF)	0	EVK	18069
48	500	0	P=2000 Inf. Dilution	lgK: -7.4 (2SF)	0	EVK	18069

49	550	0	P=2000 Inf. Dilution	lgK:-7.6(2SF)	0	EVK	18069
50	600	0	P=2000 Inf. Dilution	lgK:-7.7(2SF)	0	EVK	18069
51	25	0	Inf. Dilution	lgK:-24.3(3SF)	1	ELC	
52	25	0	Inf. Dilution	lgK:-22.(1.SD)	0	EVK	18069
53	50	0	Inf. Dilution	lgK:-19.5(1.SD)	0	EVK	18069
54	100	0	Inf. Dilution	lgK:-15.6(1.SD)	0	EVK	18069
55	150	0	Inf. Dilution	lgK:-12.9(1.SD)	0	EVK	18069
56	200	0	Inf. Dilution	lgK:-11.0(3SF)	0	EVK	18069
57	250	0	Inf. Dilution	lgK:-9.7(2SF)	0	EVK	18069
58	300	0	Inf. Dilution	lgK:-8.7(2SF)	0	EVK	18069
59	350	0	Inf. Dilution	lgK:-8.1(2SF)	0	EVK	18069

Reaction No. 75962							
S2O3-2 + NH3 + Au+1 = Au+1_NH3_S2O3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.65(4SF)	1	ELC	
2	25:30	0.1	NaClO4	lgK:20.(2SF)	0	CRV	21011

Reaction No. 75963							
Au+1_S2O3-2(2) + NH3 = Au+1_NH3_S2O3-2 + S2O3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:-4.0(0.2SD)	2	MPS	20000
Note (1): Perera and Senanayake, Inorg. Chem., 2004, 43, 3048							
2	25	0.1	NaClO4	lgK:-3.8(0.2SD)	2	MPT	20000
3	25	0.1	NaClO4	lgK:-3.5(0.2SD)	2	MVM	20000

Reaction No. 75964							
Au+1_NH3_S2O3-2 + NH3 = Au+1_NH3(2) + S2O3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:-5.4(0.4SD)	2	MPS	20000
Note (1): Perera and Senanayake, Inorg. Chem., 2004, 43, 3048							
2	25	0.1	NaClO4	lgK:-5.3(0.4SD)	2	MPT	20000

Reaction No. 75965							
Au+1 + S2O3-2 = Au+1_S2O3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:10.4(3SF)	0	EOD	21129
Note (1): Recalculated from Baranova and Ryzhenko, Geokhimiya, 1981, 7, 989							

Reaction No. 77214							
$\text{Au(s)} + 0.5\text{Cl}_2(\text{g}) = \text{Au+1_Cl-1(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:197.(3SF)kJ	4	CRV	9015

Reaction No. 77217							
$\text{Au(s)} + 2\text{Cl}_2(\text{g}) + 3\text{e-1} = \text{Au+1_Cl-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-235.14(5SF)kJ	0	CRV	9015
3	25	0	Inf. Dilution	dH:-322.2(4SF)kJ	0	CRV	9015

Reaction No. 77218							
$\text{Au(s)} + 2\text{Cl}_2(\text{g}) + 3.5\text{H}_2(\text{g}) + 1.5\text{O}_2(\text{g}) = \text{Au+3_H+1_Cl-1(4)_H}_2\text{O(3)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1192.0(5SF)kJ	0	CRV	9015

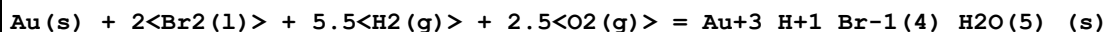
Reaction No. 77219							
$\text{Au(s)} + 2\text{Cl}_2(\text{g}) + 4.5\text{H}_2(\text{g}) + 2\text{O}_2(\text{g}) = \text{Au+3_H+1_Cl-1(4)_H}_2\text{O(4)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1489.1(5SF)kJ	0	CRV	9015

Reaction No. 77220							
$\text{Au(s)} + \text{Br}_2(\text{l}) + \text{e-1} = \text{Au+1_Br-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-115.00(5SF)kJ	0	CRV	9015
3	25	0	Inf. Dilution	dH:-128.4(4SF)kJ	2	CRV	9015

Reaction No. 77221							
$\text{Au(s)} + 2\text{Br}_2(\text{l}) + \text{e-1} = \text{Au+3_Br-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-167.3(4SF)kJ	0	CRV	9015

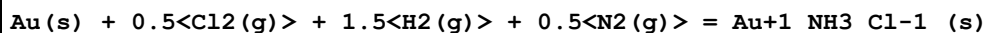
3	25	0	Inf. Dilution	dH:-191.6(4SF)kJ	2	CRV	9015
---	----	---	---------------	------------------	---	-----	------

Reaction No. 77222



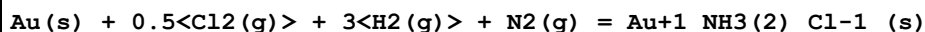
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1668.2(5SF)kJ	0	CRV	9015

Reaction No. 77223



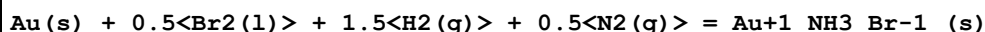
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-182.8(4SF)kJ	0	CRV	9015

Reaction No. 77224



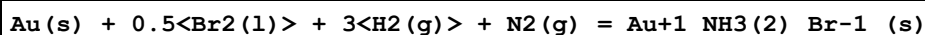
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-291.6(4SF)kJ	0	CRV	9015

Reaction No. 77225



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-154.4(4SF)kJ	0	CRV	9015

Reaction No. 77226



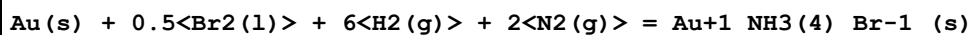
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-254.0(4SF)kJ	0	CRV	9015

Reaction No. 77227



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-333.9(4SF)kJ	0	CRV	9015

Reaction No. 77228



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-413.0(4SF)kJ	0	CRV	9015

Reaction No. 77229

Au(s) + 0.5<Br2(l)> + 9<H2(g)> + 3<N2(g)> = Au+1_NH3(6)_Br-1_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-569.9(4SF)kJ	0	CRV	9015

Reaction No. 77230							
Au(s) + 1.5<H2(g)> + 0.5<I2(s)> + 0.5<N2(g)> = Au+1_NH3_I-1_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-109.6(4SF)kJ	0	CRV	9015

Reaction No. 77231							
Au(s) + 3<H2(g)> + 0.5<I2(s)> + N2(g) = Au+1_NH3(2)_I-1_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-191.6(4SF)kJ	0	CRV	9015

Reaction No. 77232							
Au(s) + 4.5<H2(g)> + 0.5<I2(s)> + 1.5<N2(g)> = Au+1_NH3(3)_I-1_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-274.1(4SF)kJ	0	CRV	9015

Reaction No. 77233							
Au(s) + 9<H2(g)> + 0.5<I2(s)> + 3<N2(g)> = Au+1_NH3(6)_I-1_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-506.3(4SF)kJ	0	CRV	9015

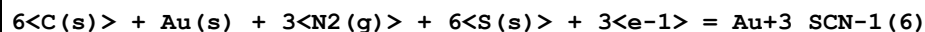
Reaction No. 77234							
2<C(s)> + Au(s) + N2(g) + 2<S(s)> + e-1 = Au+1_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:251.9(4SF)kJ	4	CRV	9015

Reaction No. 77235							
4<C(s)> + Au(s) + 2<N2(g)> + 4<S(s)> + e-1 = Au+3_SCN-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-96.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:561.6(4SF)kJ	0	CRV	9015

Reaction No. 77236							
5<C(s)> + Au(s) + 2.5<N2(g)> + 5<S(s)> + 2<e-1> = Au+3_SCN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

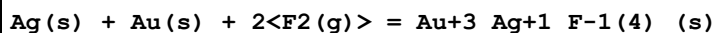
1	25	0	Inf. Dilution	dG:654.5(4SF)kJ	4	CRV	9015
---	----	---	---------------	-----------------	---	-----	------

Reaction No. 77237



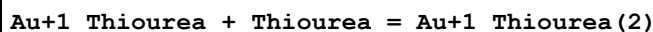
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:747.0(4SF)kJ	4	CRV	9015

Reaction No. 77238



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-641.4(4SF)kJ	0	CRV	9015

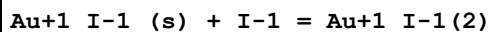
Reaction No. 77370



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.4(3SF)	1	ELC	
2	25	0.5	NaClO4	lgK:1.24(3SF)	0	EOD	22560

Note (2): Zhang & Yang, Acta Chim. Sinica, 1980, 38, 521

Reaction No. 77371



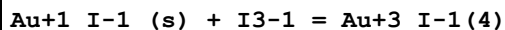
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	KNO3	lgK:0.82(2SF)	2	MIS	22560

Note (1): Erenburg & Peshchevitskii, Zh. Neorg. Khim., 1969, 14, 932 (E:485)

2	25	0	Inf. Dilution	lgK:0.85(2SF)	3	EES	
3	25	1	NaClO4	lgK:0.91(2SF)	4	MSL	22560

Note (3): Hakansson & Johansson, Chem. Scr., 1975, 7, 201

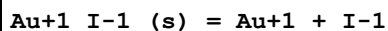
Reaction No. 77372



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.41(3SF)	4	MSL	22560

Note (1): Hakansson & Johansson, Chem. Scr., 1975, 7, 201

Reaction No. 77373



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

1	25	0	Inf. Dilution	lgK:-19.8(3SF)	1	ELC	
2	25	0.001	Unknown	lgK:-22.3(3SF)	0	MEF	22560
Note (2): Sign changed; Erenburg&Peshchevitskii, Zh.Neorg.Khim., 1969, 14, 932 (E:485)							

Reaction No. 77520							
$H+1 + 4<Cl-1> + Au+3 = Au+3_H+1_Cl-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.26(4SF)	1	ELC	

Reaction No. 77596							
$H+1 + AuO3-3 = H+1_AuO3-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	ELC	

Reaction No. 77600							
$2<H+1> + AuO3-3 = H+1(2)_AuO3-3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.7(3SF)	1	ELC	

Reaction No. 77633							
$5<OH-1> + Au+3 = Au+3_OH-1(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.6(4SF)	1	ELC	

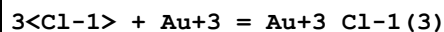
Reaction No. 77700							
$OH-1 + Cl-1 + Au+3 = Au+3_Cl-1_OH-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.5(3SF)	1	ELC	

Reaction No. 77753							
$Na+1 + 4<Cl-1> + Au+3 = Au+3_Na+1_Cl-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.26(4SF)	1	ELC	

Reaction No. 77794							
$2<Cl-1> + Au+3 = Au+3_Cl-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

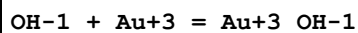
1	25	0	Inf. Dilution	lgK:15.9 (3SF)	1	ELC	
---	----	---	---------------	----------------	---	-----	--

Reaction No. 77803



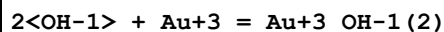
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.91 (4SF)	1	ELC	

Reaction No. 77835



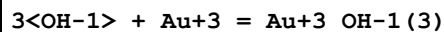
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.0 (3SF)	1	ELC	

Reaction No. 77846



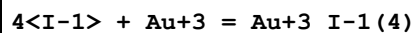
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.5 (3SF)	1	ELC	

Reaction No. 77858



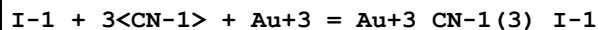
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.04 (4SF)	1	ELC	

Reaction No. 77871



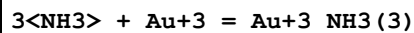
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.62 (4SF)	1	ELC	

Reaction No. 78002



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:120.2 (4SF)	1	ELC	

Reaction No. 78081



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.7 (4SF)	1	ELC	

Reaction No. 78378

6<SCN-1> + Au+3 = Au+3_SCN-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.63(4SF)	1	ELC	

Reaction No. 78399							
5<SCN-1> + Au+3 = Au+3_SCN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.59(4SF)	1	ELC	

Reaction No. 78441							
Cl-1 + 2<Br-1> + Au+3 = Au+3_Br-1(2)_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.7(3SF)	1	ELC	

Reaction No. 78442							
2<Cl-1> + Br-1 + Au+3 = Au+3_Br-1_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.17(4SF)	1	ELC	

Reaction No. 78526							
3<Br-1> + Au+3 = Au+3_Br-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.34(4SF)	1	ELC	

Reaction No. 78570							
3<CN-1> + Cl-1 + Au+3 = Au+3_Cl-1_CN-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:115.5(4SF)	1	ELC	

Reaction No. 78609							
OH-1 + Br-1 + Au+1 = Au+1_Br-1_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.9(3SF)	1	ELC	

Reaction No. 78610							
OH-1 + Cl-1 + Au+1 = Au+1_Cl-1_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.9(3SF)	1	ELC	

Reaction No. 78611							
4<SCN-1> + Au+3 = Au+3_SCN-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.15(4SF)	1	ELC	

Reaction No. 78933							
3<F-1> + Au+3 = Au+3_F-1(3)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-20.62(4SF)	1	ELC	

Reaction No. 79264							
Acetic-1 + Au+1 = Au+1_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.2(2SF)	1	ELC	

Reaction No. 79265							
2<Acetic-1> + Au+1 = Au+1_Acetic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.2(2SF)	1	ELC	

Reaction No. 79432							
OH-1 + 4<NH3> + Au+3 = Au+3_NH3(4)_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.12(4SF)	1	ELC	

Reaction No. 79852							
Au+1_SO3-2(2) + Thiourea = Au+1_Thiourea_SO3-2 + SO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:-1.2(2SF)	5	MCV	29531

Reaction No. 79853							
Au+1_SO3-2(2) + Thiourea = Au+1_Thiourea + 2<SO3-2>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:-3.6(2SF)	5	MCV	29531

Reaction No. 80417							
Au+3_Br-1_Cl-1(3) + Br-1 = Au+3_Br-1(2)_Cl-1(2) + Cl-1							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.4 (2SF)	1	ELC	
2	25	1	NaClO4	lgK:1.96 (3SF)	3	MSP	30958

Reaction No. 80418							
Au+3_Br-1(2)_Cl-1(2) + Br-1 = Au+3_Br-1(3)_Cl-1 + Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.65 (2SF)	1	ELC	
2	25	1	NaClO4	lgK:1.53 (3SF)	4	MSP	30958

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
CSM	Smith & Martell Vols 1-6 (crit.)
EBP	By Proxy
EDH	Debye-Hueckel corrections
EES	Personal estimate

EFE	Free Energy relations
EIE	Interpolation/Extrapolation (not Debye-Huekel)
EIU	Critical IUPAC review (estim.)
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
EVH	Van't Hoff Equation ($dH = \text{constant}$)
EVK	Variation of $\lg K$ with T
MCV	Cyclic voltammetry
MDG	Combination of Thermodynamic Data
MEF	Electromotoric force, not specified
MGL	Glass electrode
MIE	Current-voltage studies
MIS	Ion selective electrode
MKN	Rate of reaction
MLC	Ligand competition
MMR	Nuclear magnetic resonance
MPH	pH method, not specified
MPL	Polarography
MPS	Potentiometry and Spectrophotometry
MPT	Potentiometry
MRX	Emf with redox electrode
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference
MVM	Voltametric Methods

References

Reference No.: 5(69Hel)

Am. J. Sci., 1969, 267, 729 - 804; Helgeson HC; Thermodynamics of Hydrothermal Systems at Elevated Temperatures and Pressures

Reference No.: 8(76SmM)

Critical Stability Constants, Vol. 4, 1976 Smith RM; Martell AE; Plenum, New York

Reference No.: 140(64SiM)

Stability Constants of Metal-ion Complexes SP17, 1964 Sillen LG; Martell AE; The Chemical Soc., London

Reference No.: 552(95SMM)

NIST Critical Stability Constants of Metal Complexes Database, 1995 Smith RM; Martell AE; Motekaitis RJ; Version 2.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 600(52Lat)

The Oxidation States of the Elements, 2nd Edn, 1952 Latimer WM; and their Potentials in Aqueous Solutions; Prentice-Hall Inc., New York, USA

Reference No.: 773(85KoS)

Handbook of Chemical Equilibria in Analytical Chemistry, 1985 Kotrly S; Sucha L; Ellis Horwood Series in Anal. Chem.; Eds. Chalmers RA & Masson M; Ellis Horwood, Chichester

Reference No.: 775(71CCM)

Selected Constants, 1971 Charlot G; Collumeau A; Marchon MJC; Oxidation-Reduction Potentials of Inorganic Substances in Aqueous Solution; Butterworths, London

Reference No.: 778(78MiC)

Tables of Standard Electrode Potentials, 1978 Milazzo G; Caroli S; Wiley, N.Y.

Reference No.: 802(90Wea)

CRC Handbook of Chemistry and Physics, 70th Edn., 1990 Weast RC; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio

Reference No.: 884(72HFE)

J. Inorg. Nucl. Chem., 1972, 34, 3747 - 3751; Hancock RD; Finkelstein NP; Evers A; Stabilities of the Cyanide Complexes of the Monovalent Group IB Metal Ions in Aqueous Solution

Reference No.: 885(61BaB)

Trav. Soc. Pharm. Montpellier, 1961, 21, 111 Bardet L; Bontoux J

Reference No.: 886(62Lin)

J. Electroanal. Chem., 1962, 4, 332 - 342; Lingane J; Standard Potentials of Half-Reactions involving +1 and +3 Gold in Chloride Medium. Equilibrium Constant of the Reaction $\text{AuCl}_{14}^- + 2\text{Au} + 2\text{Cl}^- = 3\text{AuCl}_2^-$

Reference No.: 887(65PGC)

J. Chim. Phys., 1965, 62, 203 Pouradier J; Gadet MC; Chateau H

Reference No.: 888(63PKE)

Zh. Neorg. Khim., 1963, 8, 853 Peshchevitskii BI; Kazakov VP; Erenburg AM

Reference No.: 890(24JiJ)

Z. Elektrochem., 1924, 30, 286 Jirsa F; Jelinek H

Reference No.: 891(27GeR)

J. Am. Chem. Soc., 1927, 49, 1855 - 1858; Gerke RH; Rourke MD; The Potential of the Gold-Auric Oxide Electrode

Reference No.: 892(66Pou)

J. Chim. Phys. Phys. Chim. Biol., 1966, 63, 694 - 695; Pouradier J; Electrochemistry of

Gold salts. IV. Standard Potentials of the Gold-Haloaurate Systems as a Function of Temperature

Reference No.: 893(66CGP)

J. Chim. Phys., 1966, 63, 269 Chateau H; Gadet MC; Pouradier J

Reference No.: 895(48Bje)

Bull. Soc. Chim. Belg., 1948, 57, 432 - 445; Bjerrum N; Stabilite des Chlorures d'Or

Reference No.: 896(71PBK)

Russ. J. Inorg. Chem., 1971, 16, 1007 - 1009; Peshchevitskii BI; Belevantsev VI; Kurbatova NV; Hydrolysis of Chloro-complexes of Gold(III)

Reference No.: 897(72BeR)

Inorg. Nucl. Chem. Lett., 1972, 8, 849 Bekker PvanZ; Robb W

Reference No.: 898(24JiJ)

Z. Elektrochem., 1924, 30, 534 Jirsa F; Jelinek H

Reference No.: 900(90VlW)

Geochim. Cosmochim. Acta, 1990, 54, 3 - 12; Vlassopoulos D; Wood SA; Gold Speciation in Natural Waters: I. Solubility and Hydrolysis Reactions of Gold in Aqueous Solution

Reference No.: 905(95Ber)

Pure Appl. Chem., 1995, 67, 1117 - 143; Berthon G; Critical Review of Formation Constants of Selected Amino Acids with Polar Side Chains

Reference No.: 906(79Per)

IUPAC Chemical Data Series No. 22, 1979 Perrin DD; Stability Constants of Metal-Ion Complexes, First Edition Part B, Organic Ligands; Pergamon, Oxford

Reference No.: 931(71SiM)

Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 1555(91KSG)

Pure Appl. Chem., 1991, 63, 597 - 638; Kiss T; Sovago I; Gergely A; Critical Survey of Stability Constants of Complexes of Glycine

Reference No.: 1802(91Ahr)

Rev. Inorg. Chem., 1991, 11, 195 - 253; Ahrlund S; Thermodynamics of Metal Complex Formation in Solution - Some Recent Advances

Reference No.: 2392(93SKG)

Pure Appl. Chem., 1993, 65, 1029 - 1080; Sovago I; Kiss T; Gergely A; Critical Survey of the Stability Constants of Complexes of Aliphatic Amino Acids

Reference No.: 2556(93MSM)

NIST Critical Stability Constants of Metal Complexes Database, 1993 Martell AE; Smith RM; Motekaitis RJ; Version 1.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 3006(85BPJ)

Standard Potentials in Aqueous Solution, 1985 Bard AJ; Parsons R; Jordan J; Monographs in Electroanalytical Chemistry and Electrochemistry; Marcel Dekker Inc., New York and Basel

Reference No.: 3549(79ZJD)

J. Pharm. Sci., 1979, 68, 426 - 432; Zucconi TD; Janauer GE; Donahe S; Lewkowicz C;

Acid Dissociation and Metal Complex Formation Constants of Penicillamine, Cysteine, and Antiarthritic Gold Complexes at Simulated Biological Conditions

Reference No.: 4113(86BPT)

Zh. Neorg. Khim., 1986, 31, 3065 - 3068; Belevantsev VI; Peshchevitskii BI; Tselodub LD; Gold(I) Thiourea Complexes in Aqueous Solution

Reference No.: 4119(83KAP)

Zh. Neorg. Khim., 1983, 28, 2418 - 2420; Kulenov AS; Andreev AI; Pashkov GL; Kopanov AM; Belevantsev VI; Fedorov VA; Formation of Thiourea Complexes of Gold(I) in Aqueous Solutions

Reference No.: 4292(87PAS)

Talanta, 1987, 34, 111 - 121; Petrukhin OM; Andeeva EN; Shavnya YuV; Janskauskas V; Kazlauskas R; Bychkov AS; Zolotov YuA; Ion-Selective Electrodes for Gold and Silver Determination

Reference No.: 4998(86BaM)

The Hydrolysis of Cations, 1986 Baes CF Jr; Mesmer RE; Robert E. Krieger Publ. Co. Malabar, Florida

Reference No.: 5022(66BeP)

Russ. J. Inorg. Chem., 1966, 11, 836 Belyaev AV; Ptitsyn BV; The "Aging" of Rhodium Chloride Solutions

Reference No.: 5323(93DTY)

Hydrometallurgy, 1993, 32, 143 - 159; Davis A; Tran T; Young DR; Solution Chemistry of Iodide Leaching of Gold

Reference No.: 5398(82Hog)

IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 5466(72GaP)

C. R. Acad. Sci., 1972, 275, 1061 - 1064; Gadet MC; Pouradier J; Hydrolysis of Gold(I) Complexes

Reference No.: 5550(78ElG)

Acta Chem. Scand., 1978, A32, 867 - 877; Elding LI; Groning A.-B; Kinetics, Mechanism and Equilibria for Halide Substitution Processes of Chloro Bromo Complexes of Gold(III)

Reference No.: 5569(85XuO)

Metall. Trans., 1985, 16B, 455 - 463; Xue T; Osseo-Asare K; Heterogeneous Equilibria in the Au-CN-H₂O and Ag-CN-H₂O Systems

Reference No.: 5570(74SkB)

Acta Chem. Scand., 1974, A28, 740 - 746; Skibsted LH; Bjerrum J; Studies on Gold Complexes. I. Robustness, Stability and Acid Dissociation of the Tetramminegold(III) Ion

Reference No.: 5571(84MoS)

Acta Chem. Scand., 1984, A38, 23 - 29; Monsted O; Skibsted LH; Amineanionogold(III) Complexes. II. Calculation of Equilibrium Constants from Multiwavelength Spectrophotometric Measurements. Application to the Exchange of Bromide for Chloride in trans-Diaminedichloridogold(III) in Acidic Aqueous Solution

Reference No.: 5763(72SiS)

J. Inorg. Nucl. Chem., 1972, 34, 567 - 573; Singh MK; Srivastava MN; Stepwise Formation of Beryllium(II) and Aluminium(III) Chelates with Aspartic and Glutamic Acids

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6422(95BaP)

Thermochemical Data of Pure Substances, 3rd Edn., 1995 Barin I; Platzki G; Weinheim, Germany

Reference No.: 6478(81BaM)

Am. J. Sci., 1981, 281, 935 - 962; Baes CF Jr; Mesmer RE; The Thermodynamics of Cation Hydrolysis

Reference No.: 6488(88ShH)

Geochim. Cosmochim. Acta, 1988, 52, 2009 - 2036; Shock EL; Helgeson HC; Calculation of the Thermodynamic and Transport Properties of Aqueous Species at High Pressures and Temperatures: Correlation Algorithms for Ionic Species and Equation of State Predictions to 5 kb and 1000 C

Reference No.: 6495(97SSH)

Geochim. Cosmochim. Acta, 1997, 61, 1359 - 1412; Sverjensky DA; Shock EL; Helgeson HC; Prediction of the Thermodynamic Properties of Aqueous Metal Complexes to 1000C and 5kb

Reference No.: 6628(72SiS)

J. Inorg. Nucl. Chem., 1972, 34, 2067 - 2069; Singh MK; Srivastava MN; Stepwise Formation of Palladium(II), Platinum(IV), Gold(III) and Bismuth(III) Chelates with Aspartic and Glutamic Acids

Reference No.: 6684(72PeS)

Russ. J. Inorg. Chem., 1972, 17, 1386 - 1389; Peshchevitskii BI; Shamovskaya GI; Ligand Substitution Effects in Gold(III) Complexes in Methanol

Reference No.: 6686(71DuD)

Russ. J. Inorg. Chem., 1971, 16, 134 - 135; Dubinskii VI; Demidova GV; Equilibrium Constant for the Replacement of Chlorine by Iodine in the Chlorotricyanoaurate(III) Ion

Reference No.: 6758(73RLM)

Helv. Chim. Acta, 1973, 56, 2405 - 2418; Roulet R; Lan NQ; Mason WR; Fenske GP Jr; Oxidation - Reduction Reactions of Tetrachloroaurate(III) Anion with Triphenyl Derivatives of Group V Elements

Reference No.: 7275(98Atk)

Physical Chemistry, 6th Edn., 1998 Atkins PW; Oxford University Press, Oxford, U.K.

Reference No.: 7276(88CoW)

Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 7372(78BDJ)

Vogel's Textbook of Quantitative Inorganic Analysis, 1978 Bassett J; Denney RC; Jeffery GH; Mendham J; including Elementary Instrumental Analysis; Longman, London, U.K.

Reference No.: 7381(97Woo)

Physical Chemistry, 1997 Woodbury G; Brooks/Cole Publishing Co., Pacific Grove, U.S.A.

Reference No.: 7421(97Mar)

Ion Properties, 1997 Marcus Y; Marcel Dekker Inc., New York, U.S.A.

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K

Reference No.: 8875(81HKF)

Am. J. Sci., 1981, 281, 1249 - 1516; Helgeson HC; Kirkham DH; Flowers GC; Theoretical Prediction of the Thermodynamic Behavior of Aqueous Electrolytes at High Pressures and Temperatures. IV. Calculation of Activity Coefficients, Osmotic Coefficients and Apparent Molal and Standard and Relative Partial Molal Properties to 600 C and 5 Kb

Reference No.: 9015(82WEP)

J. Phys. Chem. Ref. Data, Vol. 11, Suppl. 2, 1982 Wagman DD; Evans WH; Parker VB; Schumm RH; Halow I; Bailey SM; Churney KL; Nuttall RL; The NBS Tables of Chemical Thermodynamic Properties - Selected Values for Inorganic and C1 and C2 Organic Substances in SI Units

Reference No.: 9029(97SSW)

Geochim. Cosmochim. Acta, 1997, 61, 907 - 950; Shock EL; Sassani DC; Willis M; Sverjensky DA; Inorganic Species in Geologic Fluids: Correlations among Standard Molal Thermodynamic Properties of Aqueous Ions and Hydroxide Complexes

Reference No.: 9031(97GYW)

Geochim. Cosmochim. Acta, 1997, 61, 1971 - 1983; Gammons CH; Yu Y; Williams-Jones AE; The Disproportionation of Gold(I) Chloride Complexes at 25 to 200C

Reference No.: 10174(95OHS)

J. Phys. Chem. Ref. Data, 1995, 24, 1401 - 1560; Oelkers EH; Helgeson HC; Shock EL; Sverjensky DA; Johnson JW; Pokrovskii VA; Summary of the Apparent Standard Partial Molal Gibbs Free Energies of Formation of Aqueous Species, Minerals, and Gases at Pressures 1 to 5000 Bars and Temperatures 25 to 1000C

Reference No.: 12638(99TPG)

GAMSPATH Users Manual, 1999 Talman SJ; Perkins EH; Gunter WD; A Reaction Path - Mass Transfer Program

Reference No.: 18069(03StS)

Geochim. Cosmochim. Acta, 2003, 67, 1677 - 1688; Stefansson A; Seward TM; The Hydrolysis of Gold(I) in Aqueous Solutions to 600C and 1500 bar

Reference No.: 20000(53ZLS)

Handbook of Experimental Data on Solubility Vol 1, 1953,, 672 Zdanovskii AB; Lyakhovskaya EI; Shleimovich RE; Multicomponent Water-Salt Systems. 1. Three-component Systems

Reference No.: 20001(51Zda)

Zh. Fiz. Khim., 1951, 25, 170 - 179; Zdanovskii AB; The Mechanism of Solution of Natural Salts

Reference No.: 21011(04Sen)

Hydrometallurgy, 2004, 75, 55 - 75; Senanayake G; Analysis of Reaction Kinetics, Speciation and Mechanism of Gold Leaching and Thiosulfate Oxidation by Ammoniacal Copper(II) Solutions

Reference No.: 21129(86Web)

Geochim. Cosmochim. Acta, 1986, 50, 1837 - 1845; Webster JG; The Solubility of Gold and Silver in the System Au-Ag-S-O₂-H₂O at 25C and 1 atm

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions

Reference No.: 22560(05PeP)

IUPAC Stability Constants Database, 2005 Pettit LD; Powell HKJ; Release 4.71; Academic Software, U.K.

Reference No.: 23202(09UMB)

Geochim. Cosmochim. Acta, 2009, 73, 3359 - 3380; Usher A; McPhail DC; Brugger J; A Spectrophotometric Study of Aqueous Au(III) Halide - Hydroxide Complexes at 25-80 C

Reference No.: 29531(17MKK)

J. Solution Chem., 2017, 46, 989 - 1003; Mironov IV; Kal'nyi DB; Kokovkin VV; On Gold(I) Complexes and Anodic Gold Dissolution in Sulfite-Thiourea Solutions

Reference No.: 30958(08Kud)

Talanta, 2008, 75, 380 - 384; Kudrev AG; Calculation of Equilibrium Constants by Matrix Method for Complexes of Gold(III)

Reference No.: 31585(19RVE)

J. Hazard. Mater., 2019, 380, 120725-1 - 120725-17; Ram R; Vaughan J; Etschmann B; Brugger J; The Aqueous Chemistry of Polonium (Po) in Environmental and Anthropogenic Processes